

Chapter 8

Functional Categories III — CP

In Chapters 5 and 6, we saw that the clause splits roughly into two parts: the lower part has to do with the thematic properties of the predicate (essentially the VP), and is associated with the lexical projection, while the upper part seems to have to do with non-thematic semantic properties such as mood, tense and aspect. This upper part is associated with functional projections.

In this chapter we will introduce one further clausal layer, which is higher still than T and its associated projections. This layer is also functional in nature, and bears close affinities to the T layer. However, the semantic properties associated with this higher level do not have to do with tense/aspect and other properties of the event picked out by the verb, rather they have to do with the semantic status of the whole proposition.

8.1 Complementizers

8.1.1 *that*

The evidence for the syntactic presence of this higher layer is actually fairly easy to come by. You may have noticed, whenever we have discussed the embedding of one clause inside another, that many of the examples given had an alternative. So, in addition to (1), we have (2):

- (1) I claimed she was pregnant
- (2) I claimed that she was pregnant

The word *that* in these examples optionally occurs at the start of the embedded clause. We can use simple constituency tests to show that this word forms a constituent with the following clause, rather than with the embedding verb. For example, pseudoclefting shows that *that* forms a constituent with the following TP, rather than with the preceding verb:

- (3) What she thought was that the poison was neutralised
- (4) *What she thought that was the poison was neutralised

Similarly, movement tests show that it is possible to raise *that* plus its following TP, once again suggesting they are in constituency. Interestingly, it is not possible to strand *that*, leaving it behind by moving TP on its own. The simple active sentence in (5), for example, can be passivized as in (6). When this happens, note that the *that* is carried along with the rest of the clause to the front of the sentence:

- (5) Everyone claimed that the poison was neutralized.
- (6) ?That the poison was neutralised was claimed by everyone
- (7) *The poison was neutralised was claimed that by everyone

It is also possible to move *that* plus its TP to the right, across a prepositional phrase:

- (8) It was claimed by everyone that the poison was neutralised
- (9) *It was claimed that by everyone the poison was neutralised

Verbs like *claim*, *think*, *believe*, *say*, *decide*, and many more take a clausal complement that may be introduced by the word *that* in English. Words which introduce clauses in this way are called **complementizers**. With some verbs, there is no option: the complementizer is obligatory:

- (10) *Jason whispered the phoenix had escaped
- (11) Jason whispered that the phoenix had escaped

In other languages, complementizers are obligatory for all clausal complements:

- (12) Thuairt mi gu bheil i tinn (Gaelic)
Say-[PAST] I that was she ill
'I said that she was ill.'
- (13) *Thuairt mi bheil i tinn (Gaelic)
Say-[PAST] I that was she ill
'I said that she was ill.'
- (14) J'ai dit qu' elle était malade (French)
I-have said that she was ill
'I said that she was ill.'
- (15) *J'ai dit elle était malade (French)
I-have said she was ill
'I said she was ill.'

The appearance of the word *that* in English in constituency with the embedded clause motivates the idea that it is a syntactic element which merges with TP, but does not tell us anything about the category of this element. An obvious possibility is that the relevant category is D, since, in English, the word *that* also serves as a demonstrative. However, this approach fails for a number of reasons. Firstly, no other demonstrative can occur in this position with the appropriate reading:

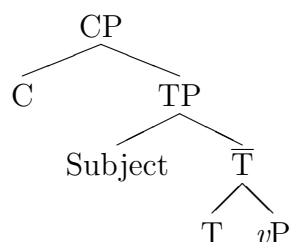
(16) *I said this he left

(17) *Jason knew those Medea had cast the spell

This strongly suggests that the word *that* is not a demonstrative in these contexts; secondly, other languages have complementizer elements that are nothing like any element of their demonstrative system. As can be seen from the examples above, in French, the complementizer is *que*. However, the demonstratives are *ce/cela/ça*; similarly, in Gaelic the demonstratives are *seo/sin*, but the complementizer is *gu*. The third argument against treating *that* as a D is that there are other complementizers in English, which look nothing like any part of the determiner system. We will look at these in more detail below.

For these reasons, most linguists analyse the word *that*, and other complementizers, as elements of a new syntactic category, C (for Complementizer). C merges with TP and projects, giving structures like (18):

(18)



8.1.2 Declaratives and Interrogatives

It is clear that C must be a functional rather than a lexical category, as it does not assign a θ -role. The semantic function of complementizers is a complex matter, but intuitively, a complementizer indicates how the hearer should think of the proposition expressed by its clause: the main two possibilities are whether the clause should be thought of as a simple statement of fact, or a question about the facts. So we find complementizers introducing questions, as well as the cases with simple propositions that we have already met:

(19) Jason asked **whether** the potion was ready

(20) Medea wondered **if** the potion was ready

We can see that *whether* and *if* are in complementary distribution to *that* in English:

(21) *Jason asked whether that the potion was ready

(22) *Medea wondered if that the potion was ready

and that they are in constituency with their following clause:

(23) What Jason asked was whether the potion was ready

(24) *What Jason asked whether was the potion was ready

(25) ?What Medea wondered was if the potion was ready

- (26) *What Medea wondered if was the potion was ready

This suggests that *if/whether* are also complementizers. Given their different semantic function, we can distinguish *that* from *whether/if* by means of an interpretable syntactic feature. The common term used for clauses which are questions is **interrogative** while clauses which are non-questions are said to be **declarative**. The status of a clause as interrogative or declarative is its **clause type**.

We will assume, then, a feature [clause-type] whose values are either [Q] (for an interrogative clause) or [Decl] for a declarative. Q is, of course, a mnemonic for question. These features are interpretable on C, and give rise to the different semantics of declaratives and questions.

We can state the c-selectional requirements of predicates which take a clausal complement in the lexical entries for the predicates as follows:

- (27) ask, [V, *uC*]

- (28) exclaim, [V, *uC*]

However, we still need to distinguish between these two classes of predicates, since the predicate seems to be able to determine whether its complement is interrogative or declarative:

- (29) *Ron asked that the potion was ready

- (30) *Medea exclaimed if the potion was ready

(Note: this last example is well-formed for some speakers on a reading where, whenever the potion was ready, Medea exclaimed. This reading is irrelevant here, and arises from a different syntactic structure, which we will cover in section 8.3 below).

The question is whether this information should be built into the c-selectional properties of the selecting predicate, or whether it is a simple s-selectional fact.

There is some evidence that the semantic (s-selectional) approach is the correct one. There are verbs like *maintain* which behave just like *exclaim* in rejecting a [Q] complement clause:

- (31) Julie maintained that the barman was sober.

- (32) *Julie maintained if the barman was sober.

These verbs, however, also allow DP complements:

- (33) Julie maintained her own ideas over the course of the argument.

Interestingly, these verbs seem to be incapable of combining with a DP whose semantics is that they refer to a question:

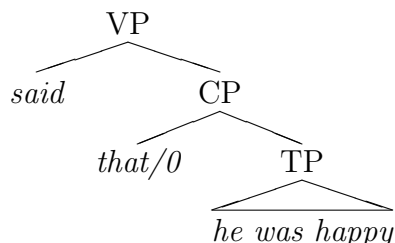
- (34) *Julie maintained her own questions over the course of the argument.

Since, from a syntactic point of view *her own ideas* and *her own questions* are identical, we need to appeal to a semantic incompatibility to rule out (34). Since we need to appeal to this semantic explanation in other cases, it makes sense to appeal to it to rule out examples

like (30) as well.

Given these lexical entries for verbs that select a declarative or interrogative clausal complement, we are led to assume that the cases with an optional *that* are actually cases with a zero complementizer, so that the higher predicate always selects a CP.

(35)



This position naturally leads to another question: do simple matrix sentences contain complementizers? That is, should we assume that a matrix sentence is really a CP rather than a TP?

There is no agreed upon answer to this question. Some linguists have argued that, in the interests of economy, a simple sentence is just a TP, since there is no evidence for more structure. Other linguists have argued, on the basis of uniformity across clauses, that simple sentences must be CP, since embedded sentences clearly are.

It is certainly the case that some simple sentences are CPs. We can see this with matrix questions, like the following:

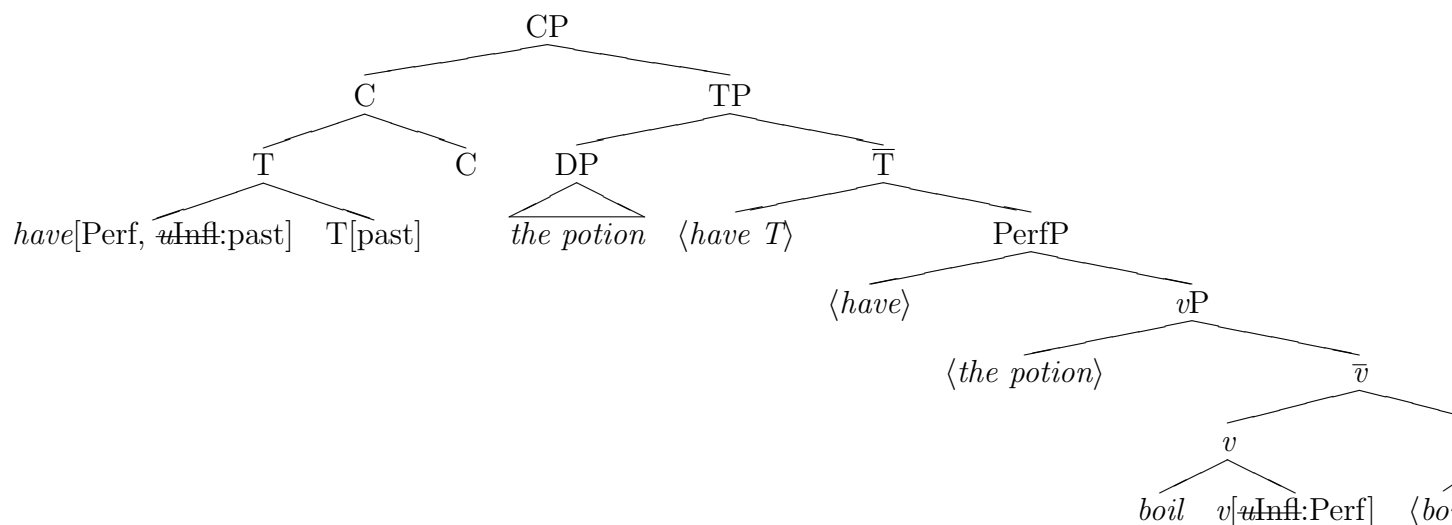
(36) Had the potion boiled over?

(37) Did the magic work?

These simple interrogatives (often termed **yes/no questions**, because of their most obvious answers) contrast with their declarative counterparts in an obvious way. The auxiliary appears, not after the subject, but before it. Traditionally, this construction is known as **Subject-Auxiliary Inversion (SAI)**, since the first auxiliary inverts with the subject (we discussed this construction briefly in Chapter 1).

What is the analysis of such structures? If we assume that matrix sentences are CPs, then we have a simple analysis: the collection of features that projects T (i.e., tense) moves into the C position, in much the same way as the highest auxiliary moves into T in modern English and as the verbal cluster moves into T in French (see the discussion in Chapter 5). We will assume that tense adjoins to C, in the same way as we assumed that auxiliaries adjoined to T.

(38)



What we have here is a situation where the perfect auxiliary is moved to T, creating an adjunction structure with T. Tense has then raised into C, again creating an adjunction structure. The resulting structure is created by successive steps of head movement.

Movement of the perfect auxiliary to T is triggered by the strength of the valued tense features in *have*, as discussed in Chapter 5. The question that remains is why the complex in T moves to C.

Once again, to motivate movement, we appeal to strength of features. If we assume that T bears an uninterpretable unvalued clause type feature [*u*clause-type:], then this feature can check with a [Q] feature on matrix C. On the assumption that the clause type feature on T when valued as Q is strong in English, T will move to C, carrying along everything inside it.

Lets see how this works. At the point in the derivation where Perf has already moved to T, we have the following rough structure:

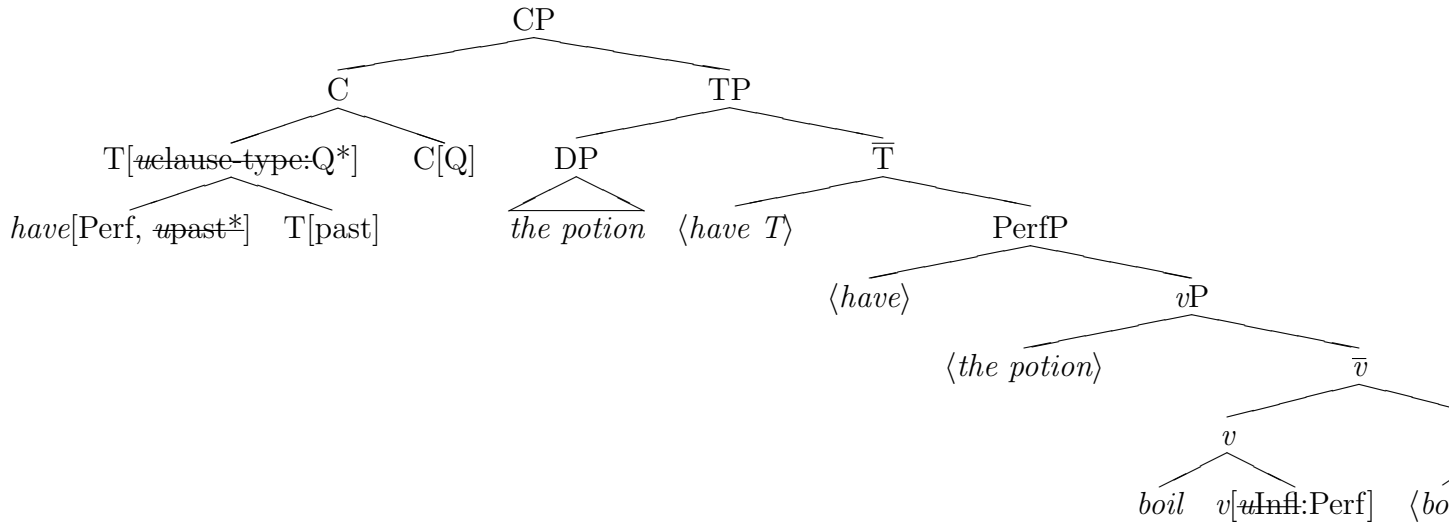
(39) [_{TP} Subject Perf+T[*u*clause-type:] ...

When C[Q] is Merged with this TP, it values the clause type feature of T as strong Q:

(40) C[Q] [_{TP} Subject Perf+T[*u*clause-type:Q*] ...

Given that Q on T is strong, it raises into a local relation with the feature that it Agrees with, giving (41) as the final structure:

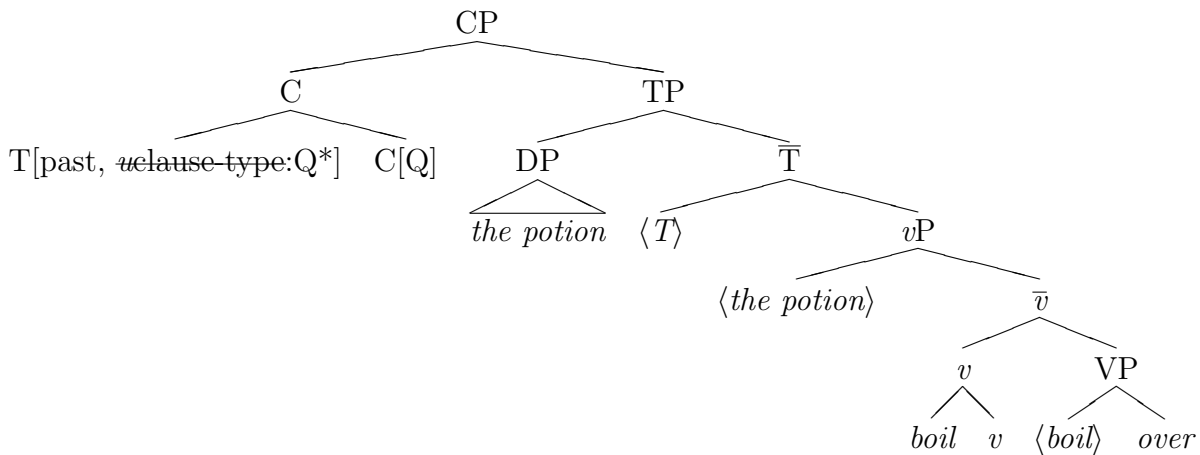
(41)



Interestingly, this analysis also predicts the correct occurrence of *do*-support in these sentences. Interrogative C triggers movement of T to C. Once this movement has taken place, T no longer c-commands little *v*. This means that a chain cannot be formed between them, and so the Rule for Pronouncing Tense cannot apply; *do*-support applies as a last resort instead.

(42) Did the potion boil over?

(43)

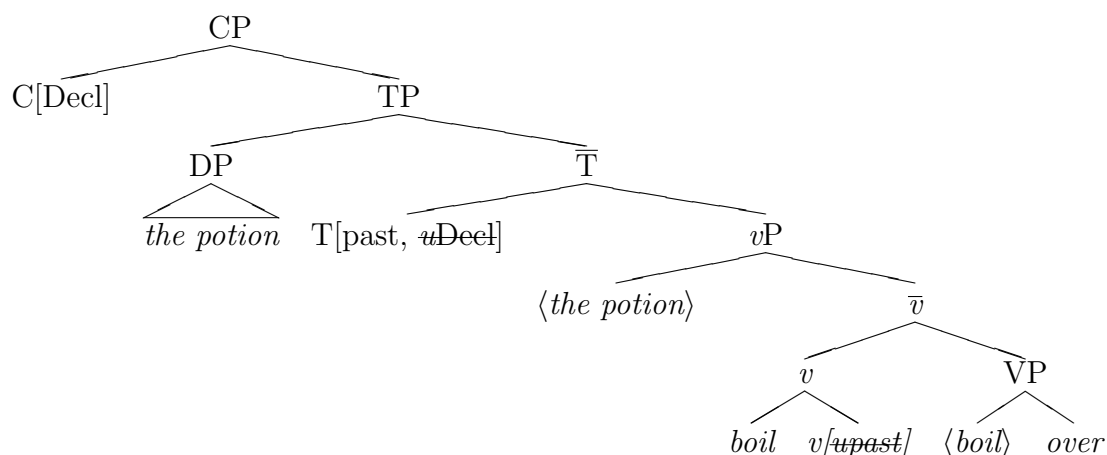


The existence of SAI structures strongly suggests that a CP layer is present in at least matrix yes/no questions.

We will assume further that CP layers are actually present in all matrix sentences, but that, for matrix declaratives, the topmost C position is occupied by a phonologically null complementizer, which values the uninterpretable clause type feature on T as Decl. The difference between Q and Decl is that the latter is weak, and so no movement takes place. The structure of a simple clause is then the following:

(44) The potion boiled over?

(45)



In future structures we'll usually abbreviate clause type features as follows:

- (46) a. [clause-type:Q] (interpretable clause type feature with value Q) abbreviated as [Q]
 b. [*u*clause-type:] (unchecked, unvalued uninterpretable clause type feature) abbreviated as [*u*clause-type]
 c. [*u*clause-type:Q*] (checked uninterpretable clause type feature valued as strong Q) abbreviated as [Q*].

These abbreviations are analogous to the ones we developed for case.

8.1.3 Subject Clauses

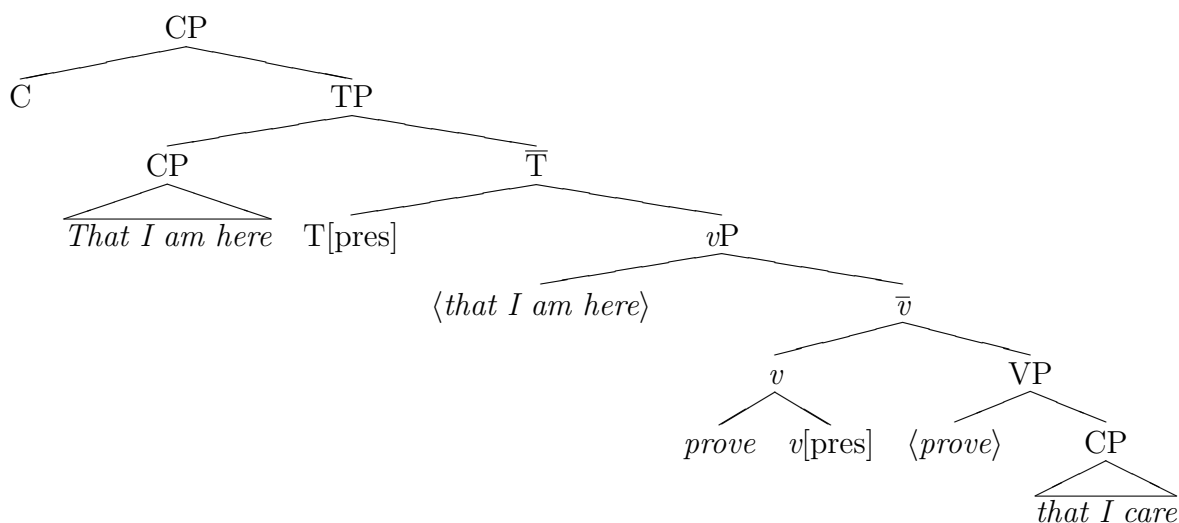
The examples we have seen so far of clauses introduced by complementizers have all been in complement position. We also, however, find clauses in subject positions:

- (47) [That Jason abandoned Medea] stinks!
 (48) [That the golden thread would show Jason his path through the labyrinth] was obvious.
 (49) [That I am here] proves [that I care].
 (50) [That Zeus was so promiscuous] astounded the other gods.
 (51) [That Medea murdered Jason] didn't surprise anyone.

The first two examples here are cases of one place predicates: *stink* and *be obvious*. Example (49) is a case where a two place predicate allows both of its arguments to be clausal, while examples (50) and (51) show cases where we have a clausal subject but a non-clausal object.

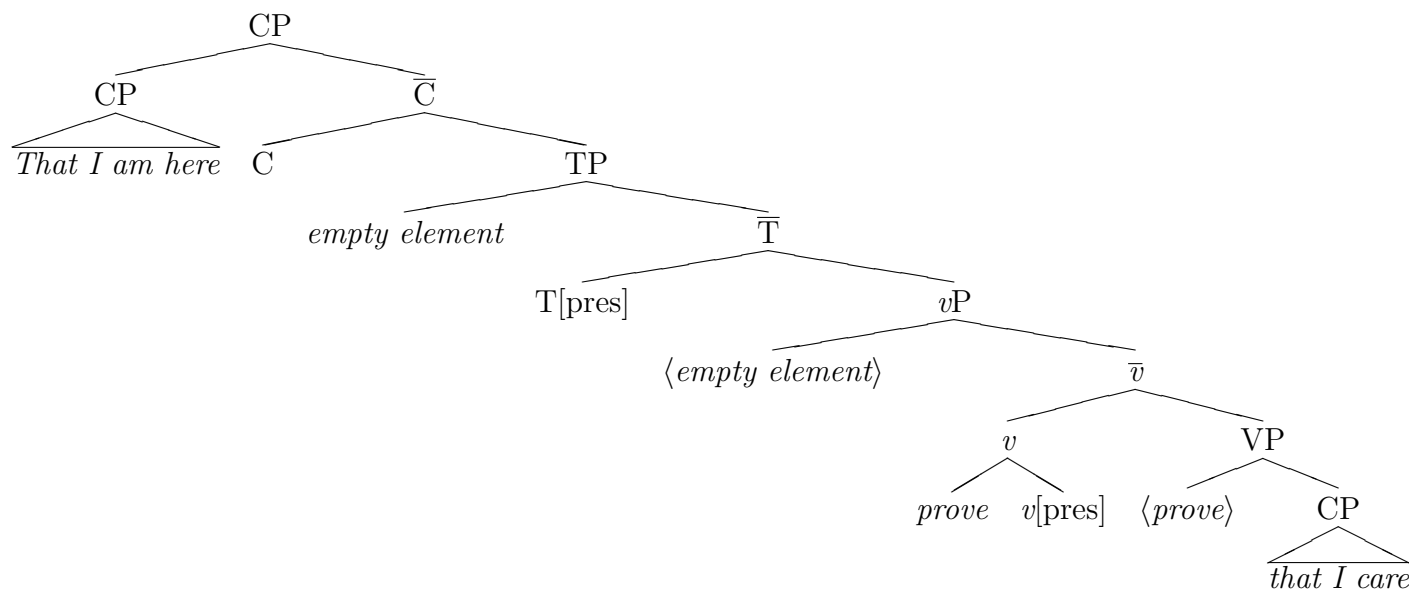
A natural analysis of these examples is to assume that the clausal subject is in the specifier of TP position, and that it has raised from a VP internal position. So an example like (49) would have the structure in (52):

(52)



An alternative analysis that has been proposed treats the first clause as being in the specifier of CP, with an empty element of some kind in the subject position. The empty element is the true subject, and the CP is associated with it semantically:

(53)



In order not to prejudge the analysis, let us call the CPs we are interested in here **initial CPs**

The reason that the second analysis for initial CPs has been proposed is that these CPs do not behave like other subjects. For example, they do not appear to undergo Subject Auxiliary Inversion:

(54) *Did that Medea killed her children upset Jason?

(55) *Has that we have arrived back at our starting point proved that the world is round?

Similarly, while it is clearly possible to have a DP subject in an initial CP, it is not possible to have an initial CP in an initial CP:

- (56) [That [the answer] is obvious] upset Hermes.
 (57) *[That [that the world is round] is obvious] upset Hermes.
 (58) *[That [whether the world is round] is unknown] bothered Athena.

These examples show that ‘normal’ DP subjects have a different distribution to the initial CPs that we are interested in, suggesting that they are in different syntactic positions.

Finally, in other languages, it is clear that an initial CP is not in subject position, because a special pronoun form appears. The following example is from Dutch:

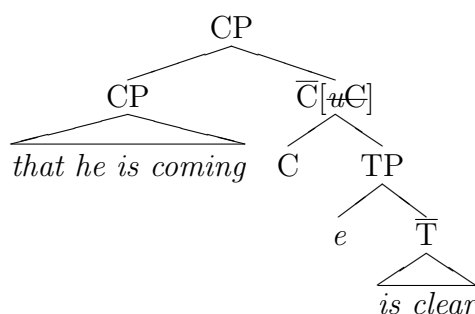
- (59) Dat hij kommt, dat is duidelijk
 that he comes that is clear
 ‘That he will come is clear.’

The idea, under the second analysis, is that Dutch and English are structurally just the same, but that the English equivalent of *dat* is null (represented here just as *e*):

- (60) That he is coming *e* is clear.

Let us think about how this proposal would actually work. The null pronoun Merges with the bar-level projection of little *v* and then raises to the specifier of TP, checking case and the EPP feature of T. We then Merge C, and then a previously constructed CP. This Merger needs to be triggered by a selectional feature of C, so the matrix C itself has to have a *uC* feature which is projected to $\bar{C}$ and then checked by the initial CP:

- (61)



One theoretical advantage that this proposal has is that it allows us to maintain our idea that the EPP feature on T is an uninterpretable [D] feature of T. This feature is satisfied by merging into the thematic subject position the demonstrative pronoun *dat* in Dutch and raising it to the specifier of TP to satisfy the EPP feature. The derivation is just the same for English, except that the pronoun is null. If we were to adopt the alternative analysis, and to put the initial CP in the specifier of TP, we would need to rethink exactly what the EPP feature is, since it would then be able to be satisfied by DP or CP.

Under the null pronoun analysis, we also have a way of encoding the fact that SAI does not occur with initial CPs. Recall that SAI is triggered by C valuing the clause type feature

of T as [uQ^*]. We can block SAI occurring where we have an initial CP by simply stating that the matrix C[uC] that appears in initial CP constructions is simply incompatible with the Q feature. Since this special C is needed to have an initial CP, initial CPs are incompatible with SAI. Of course, without independent motivation, this is just a way of formalizing our description, rather than providing an explanation. It does, however, at least allow us to capture the correct patterns.

Finally, the proposal makes possible an analysis which predicts why initial CPs can't occur inside other initial CPs. If we look at an initial CP with a DP subject, the subject occurs after the complementizer:

(62) [That **Jason** arrived] infuriated Medea.

(63) [Whether **Agamemnon** had triumphed] was unknown.

Under our new analysis of initial CPs, however, they are in the specifier of a null C[uC], and the subject position is filled by an empty element. This explains why we never find an initial CP after a complementizer: initial CPs are themselves actually Merged with the projection of a special complementizer, so we will never find them following *that* or *whether*. However, the proposal makes mysterious the unacceptability of examples like the following:

(64) *[[That Jason had arrived] C *e* was obvious] C *e* infuriated Medea.

If we say that the null C[uC] is restricted to matrix clause positions, then (64) will be ruled out, since there is an occurrence of this null C in an embedded clause. This same restriction will correctly rule out examples like the following:

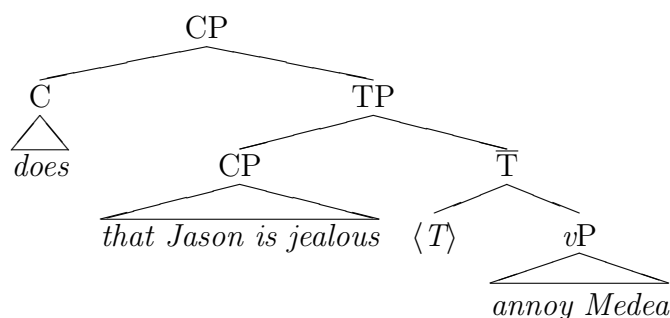
(65) *I said that that Jason was jealous annoyed Medea

We have adopted a syntactic approach to the behaviour of initial subjects which assumes that they are introduced by a special null C, which bears a selectional C feature, allowing a CP to merge in its specifier. We adopted this analysis because it allowed us to deal with certain differences between initial CPs and DP subjects by claiming that there was a structural difference between them. An alternative approach would be to adopt the idea that initial CPs really are in the standard structural subject position (the specifier of TP) and to say that their odd syntax is the result of something else.

One of the earliest approaches to initial subjects adopted this second idea. Ross, in his PhD thesis, argued that the odd syntax of initial CPs was due to a general parsing principle which prohibited CPs from being **internal** to other CPs, where a constituent is internal to another constituent just in case it is surrounded by phonological material from that other constituent. He termed this the **Internal S Condition**.

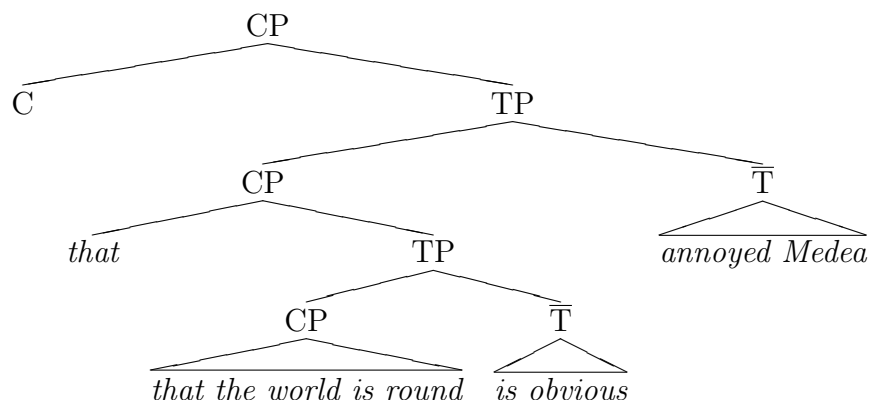
Let us see how this would work: firstly we would assume an analysis where initial CPs are in the specifier of TP. If T to C movement takes place, then the initial CP is internal to the matrix CP, since there is phonological material from the matrix CP on both sides of the initial CP:

(66)



Similarly, if an initial CP is the subject of another initial CP, then clearly it will be internal. On one side it will have the higher CPs complementizer, and on the other, the rest of the sentence:

(67)



However, there are sentences which seem to fall foul of the Internal S Condition, but which are perfectly well formed:

(68) Believing [that the world is flat] gives one some solace.

In (68) the matrix CP contains *believing* and *gives one some solace*, which appear on either side of the internal CP. This should be ruled out by the Internal S Condition, but the sentence is, of course, perfectly grammatical.

For this reason, we will adopt the specifier of CP analysis for internal CPs from now on. We should note, though, that this analysis itself has some problems. The most worrying is how to rule out the following:

(69) *e* is obvious.

We will return to this problem in section 8.2.8.

8.2 Non-finite CP Complementation

In this section we will explore the behaviour of non-finite clausal complements to various kinds of predicates. Before we do this, we will tease out an implication of the ideas we

developed about θ -roles in Chapter 4. This idea will be important for our discussion of clausal complementation.

Recall that lexical items are specified with c-selectional features, and that these features are associated with θ -roles in the lexical representation of words. Since all uninterpretable features have to be checked, it follows that all the θ -roles of a particular predicate will be assigned.

In the theory we developed in Chapter 4, this was the only way that something could be interpreted as an argument. The UTAH specified that certain syntactic configurations had certain thematic interpretations. If a DP, say, was not in one of these configurations, then it could not be interpreted as an argument.

This theory made sense of expletive elements, which are not interpreted as arguments, and which are inserted just to satisfy EPP requirements. Expletives are inserted in positions where no θ -role is assigned (the specifier of TP, for instance). In a sentence like (70), all the θ -roles are assigned internal to the v P projected by unaccusative *arrive*:

(70) There arrived a new actor.

Although these sentences are fairly marginal in English, they are possible. What is impossible is to put a non-expletive DP in the place of the expletive:

(71) *The doctor arrived a new actor.

It follows, then, from our treatment of the relationship between the syntax and the semantics of arguments, that a DP can never be interpreted as an argument unless it has Merged in a position which the UTAH determines as an **argument position**.

A second point about the theory of θ -roles which we need to remember is that a DP is assigned only a single θ -role. An argument receives its θ -role in the position that it is Merged. Once it has a θ -role, it may not move into another position where it will receive a second role. This restriction is what rules out examples like the following:

(72) *Medea murdered

If a single DP were able to receive two θ roles, then *Medea* could be Merged as sister to *murder*, thus receiving a Theme interpretation, and then raise to the specifier of little v P, where it would receive an Agent interpretation. However, this is not possible, and we will assume that this is because DPs may occur in only one **argument-position** in a derivation.

Let us sum up these ideas as follows:

- (73) a. A DP may receive only one θ -role.
b. A DP must receive a θ -role.

We will use these ideas to develop an analysis of infinitival complementation. The analysis we will develop is only as strong as these generalisations, so that, if it turns out that these generalisations have to be jettisoned, our analysis will have to be reconsidered. See the reference section for suggestions for alternative approaches to infinitival complementation that reject these ideas from θ -theory.

8.2.1 PRO and Control Clauses

We have seen that finite CPs may act as subjects and as complements to verbs. We met examples of non-finite clausal complements in Chapter 5, when we used the distribution of non-finite *to* to motivate TP:

- (74) Medea tried [to poison her children].
 (75) Odysseus planned [to hear the sirens].

Note that the verbs in the embedded clauses are two place predicates. The Agent of these verbs is interpreted as being referentially dependent for its interpretation on the Agent of the matrix verb; so in (74), Medea, not anyone else, is the person envisaged as poisoning her children. Likewise, in (75), it is Odysseus who is going to hear the sirens.

Note that in neither of these examples is it possible to realise the subject of the lower clause overtly:

- (76) *Medea tried [Medea to poison her children].
 (77) *Medea tried [the nurse to poison her children].
 (78) *Odysseus attempted [Odysseus to hear the sirens].
 (79) *Odysseus attempted [the helmsman to hear the sirens].

Poison is a two place predicate in (74), assigning an Agent and a Theme θ -role. The Theme θ -role is assigned to *her children*. What receives the Agent θ -role? It can't be *Medea* in the matrix clause, because Medea has a θ -role which it receives from *tried*. This means that, under the assumptions we have developed so far, there must be a null element in (74) which is Merged in the specifier of little *v*. This null element receives a θ -role from the verb in the non-finite clause. This whole clause then gets a θ -role from the matrix verb. For convenience, we term this null subject PRO. Given our theory so far, PRO is Merged in the specifier of little *v* (we will address the question of whether PRO moves below):

- (80) Medea tried [to PRO poison her children].

In sentences like (80), PRO is said to be **controlled** by the matrix subject. Clauses with PRO in them are then said to be **Control Clauses**.

PRO may also occur in structures where it is not controlled by another DP, in which case it is arbitrary in its interpretation, a little like the pronoun *one* in formal English:

- (81) PRO to steal hexes from witches is dangerous
 (82) If one were to steal hexes from witches, then that would be dangerous.
 (83) It is not allowed to PRO perjure oneself.

We will not delve into the question of when PRO gets a controlled interpretation, and when it gets an arbitrary interpretation here, but we will try instead to solve the syntactic question of what PRO's distribution is.

8.2.2 ECM clauses

Control clauses have PRO as their subject, and the control clauses we have seen so far do not allow overt DP subjects. However, there are non-finite complement clauses which allow overt subjects:

(84) We believed Zeus to be omnipotent.

(85) No one expected Agamemnon to win.

Each of the matrix verbs in these examples is also a 2 place predicate relating an individual to a proposition. We can see this by using finite clauses instead of non-finite ones:

(86) We believed that Zeus was omnipotent.

(87) No one expected that Agamemnon would win.

On the assumption that the θ -role assigning properties of these verbs remains constant, this suggests that the structure of these sentences is as follows:

(88) We believed [Zeus to be omnipotent]

(89) No one expected [Agamemnon to to win]

If we substitute pronouns in for the subjects of these non-finite embedded clauses, we find that they are accusative:

(90) We believed [him to be omnipotent]

(91) No one expected [him to to win].

The traditional name for these constructions is **ECM constructions**, where ECM stands for Exceptional Case Marking. This name is really just a leftover from an earlier theory of this phenomenon (beforehand they were referred to as Raising-to-Object constructions). We shall keep this name, even though the theory we develop of this phenomenon will turn out to involve just the same case assigning mechanism as is at play in standard verb object structures.

So in English we have seen so far two different types of non-finite clause: one type disallows overt subjects, and requires PRO. This is what we see with verbs like *try*, *attempt*, *endeavour*. The other type allows overt subjects in the non-finite clause, as seen above. In fact, matters turn out to be more complicated because some of the verbs that allow an overt subject prohibit PRO, while others don't:

(92) *We believed to PRO be omnipotent.

(93) No one expected to PRO win.

The question we are faced with then, is how to syntactically distinguish the three types of verb we have just met: verbs that take non-finite complements with obligatory PRO (*try*, *attempt*, *endeavour*); verbs that take non-finite complements but don't allow a PRO subject in these complements (*believe*, *consider*, *think*, *assume*); and verbs that allow both PRO and overt subjects (*want*, *expect*, *prefer*). These verbs clearly fall into different semantic classes,

but we'd like to know how their syntactic specification differs. We will attempt to solve this problem by developing the account of Case and EPP that we have used so far.

8.2.3 C in Non-finite Clauses

Given that finite clauses can be headed by complementizers, the natural assumption is that non-finite clauses may be as well, completing the parallel. In fact there is clear evidence that this is the case:

- (94) I intended [for Jenny to be present].
 (95) [For you to do that] would be a mistake.

In (94), we see that we have an overt subject in the non-finite clause (*Jenny*) which is preceded by an element which looks like the preposition *for*.

At first blush, a plausible approach to this sentence would be to treat *for Jenny* as a constituent: a simple PP.

- (96) I intended [for Jenny] [to be present].

This would predict that it should be possible to cleft this PP, or to topicalise it:

- (97) It was for Jenny that I intended to be present.
 (98) For Jenny, I intended to be present.

Notice that these sentences are in fact well formed, but they do not have the meaning we would expect on the basis of our original sentence. In each of these cases Jenny is treated as a **benefactive** argument: someone for whose benefit either the act of intending took place (unlikely, but possible), or the act of being present took place. The most natural interpretation for the person who is present is the subject (*I*), rather than Jenny.

Now, while this is a possible interpretation for our original sentence, it is not the only one. The interpretation we are interested in here is one where Jenny is the person who is present, and what I have intended is that this situation should come about.

This means that our original sentence is actually structurally ambiguous. In one structure, *for Jenny* is a PP which is interpreted as holding a benefactive relation with the verb *intend*. In the other structure, *Jenny* is interpreted as being the subject of the clause.

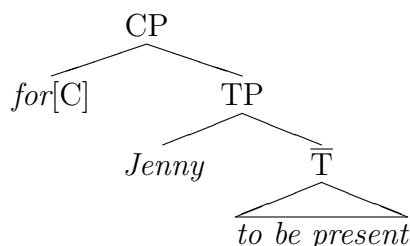
Further evidence that the *for Jenny* string in (97) is part of the embedded clause, rather than a separate constituent of the matrix clause, can be found by looking at pseudoclefting:

- (99) What I arranged was for Jenny to be present.
 (100) *What I arranged for Jenny was to be present.

The example in (100) is ungrammatical on the reading where the arrangement involved Jenny being present, rather than the speaker being present. There is again an alternative reading where (100) is grammatical, but this is irrelevant for our purposes (this is the reading where *Jenny* behaves as a benefactive argument of *arrange*).

We can see, then, that distributionally, *for* in non-finite clauses occurs immediately before the subject of the clause, and forms a constituent with CP. This is exactly parallel to *that* in finite clauses, so we have good reasons to classify *for* as a complementiser.

(101)



Interestingly, whenever the complementizer *for* appears in a non-finite clause, the clause must have an accusative subject, much like the obligatory ECM verbs we met in the last section. PRO is ruled out:

(102) Jason intended for him to learn magic.

(103) *Jason intended for he to learn magic.

(104) *Jason intended for PRO to learn magic.

The same observation can be made for this non-finite C in subject clauses:

(105) *For to do that would be a mistake.

It appears that we need a way of letting C determine the case of the subject of the non-finite clause. This is straightforward enough, since we simply extend the mechanism of case checking that we have already developed. Our treatment of case allows a case checking relation to be established between a higher functional head and the closest DP it c-commands. Assuming that the non-finite T that *for* selects has no case feature at all, then a case checking relation can be established between *for* and the subject of the clause. To see this, look at the following schematic structure at a point in the derivation where *v* has been Merged:

(106) $v[\text{acc}] \text{ V DP}[3, \text{sing}, \text{fem}, \text{case}]$

The [acc] feature on *v* will value the [case] feature on the pronominal object:

(107) $v[\text{æe}] \text{ V DP}[3, \text{sing}, \text{masc}, \text{æe}]$

We then Merge the subject (also a pronominal), followed by T and the complementizer *for*:

(108) $[_{CP} \text{ for}[\text{acc}]... \text{ T}... \text{ DP}[3, \text{sing}, \text{masc}, \text{case}] \text{ } v[\text{æe}] \text{ V DP}[3, \text{sing}, \text{masc}, \text{æe}]$

Now, just in the same way as *v* valued the [case] feature of the object, *for* will value the [case] feature of the subject:

(109) $[_{CP} \text{ for}[\text{æe}]... \text{ T}... \text{ DP}[3, \text{sing}, \text{masc}, \text{æe}] \text{ } v[\text{æe}] \text{ V DP}[3, \text{sing}, \text{masc}, \text{æe}]$

This structure will eventually have the following Spellout:

- (110) $[_{CP} \text{ for}[_{a\acute{e}e}]... T... DP[3, \text{sing}, \text{masc}, a\acute{e}e] v[_{a\acute{e}e}] V DP[3, \text{sing}, \text{masc}, a\acute{e}e]]$
 for to him see her

This means that we get a non-finite clause with two accusatives:

- (111) I arranged [for him to see her].

Notice, though, that the word order generated by the structure in (110) is not right. The non-finite T precedes rather than follows the accusative subject. However, this also follows from our treatment of the syntax of T and how it relates to the subject (see Chapter 6): we adopted the idea that case features on T were divorced from the EPP feature of T, and that it was the EPP feature that drove movement. If non-finite T differs from finite T just in having no case feature, then it is the EPP feature of T in (110), that causes *him* to raise to its specifier, deriving the correct order.

- (112) $[_{CP} \text{ for}[_{a\acute{e}e}] [DP[_{a\acute{e}e}] T[_{uD^*}]... \langle \text{him} \rangle v[_{a\acute{e}e}] V DP[_{a\acute{e}e}]]]$
 for him to $\langle \text{him} \rangle$ see her

What this analysis does then, is capitalize on the distinction we drew between the EPP feature of T and its case feature. Non finite T lacks a case feature, so the case of the subject in a non-finite clause is determined by something else. However, the word order is still determined by the EPP property of T.

8.2.4 Null case and PRO

We can now return to the phenomenon of control clauses and provide an answer to the following question: what mechanism ensures that control clauses selected by verbs in the *try* class have a PRO in them. The intuition we will pursue here is that, just as case is the feature that regulates the appearance of an accusative DP in *for* clauses, it is case that regulates the appearance of PRO in control clauses.

We therefore extend our case features for English to include, as well as [nom], [acc] and [gen], a new feature. We will call this case feature [null], and its morphological effect is to ensure that the pronoun that it is on is unpronounced (just as the morphological effect of [nom] on [3, sing, masc] is to ensure it is pronounced *he*).

We now propose that a verb like *try* selects a CP headed by a complementizer with this [null] feature. This complementiser is itself not pronounced.

Our assumption leads to the following derivation for a control clause: First the little *vP* is constructed, with a pronoun in its specifier:

- (113) $[_{vP} \text{ pronoun } v[_{a\acute{e}e}] V \text{ Object}[_{a\acute{e}e}]]$

Then non finite T *to* is Merged. T has an EPP feature but no case feature. The EPP feature on T and the D feature of the pronoun in the specifier of little *vP* agree, and the latter raises to the specifier of TP:

- (114) $\text{pronoun } T[_{uD^*}] \langle \text{pronoun} \rangle v[_{a\acute{e}e}] V \text{ Object}[_{a\acute{e}e}]$

Null C is merged, bearing the case feature [null]. This values the case feature on the pronoun as [null]

(115) C[~~null~~] pronoun[~~null~~] T[~~uD*~~] ⟨pronoun⟩ v[~~acc~~] V Object[~~acc~~]

The selecting verb *try* is Merged as is the remainder of the structure. Once this structure goes through the rules of spellout, the pronoun bearing [null] is spelled out as just a null element (or alternatively, it is not spelled out).

(116) try [C[~~null~~] pronoun[~~null~~] T[~~uD*~~] ⟨pronoun⟩ v[~~acc~~] V Object[~~acc~~]
try PRO to ⟨PRO⟩ see him

Note that we have an EPP driven movement of the pronominal feature bundle which is eventually spelled out as PRO.

This analysis immediately accounts for the impossibility of examples like the following:

(117) *Medea tried her to leave.

In order to ensure that no DP can occur as the subject of one of these structures, we simply state that the [null] feature is only a feature of pronouns. If we have a full DP Merged as the subject of the non-finite clause, then its case feature will not be checked by the null C:

(118) *Medea tried 0[null] Jason[case] to leave

This system also captures the fact that non-finite clauses with *for* must have an overt accusative subject. If we Merge in a pronoun, whose case feature is valued by *for*, then it will have an overt spellout as an accusative pronoun. Examples like (119) simply can't be generated, because the case value of any pronominal subject would be [acc] and not [null]:

(119) *I arranged for PRO to see her.

We have now provided a partial answer to the question of the distribution of subjects of non-finite clauses. The answer rests on the idea that T in non-finite clauses actually does not check case. Rather, the case of the subject of an infinitival clause is checked by some other element. In both of the situations we have examined here, that element is the C which selects the non-finite T. The actual position of the subject is determined by the EPP feature on non-finite T, rather than the case feature, just as in our revised analysis of finite clauses.

8.2.5 Cross-clausal case checking and ECM

How do we deal with those cases where there is no overt complementizer but where the embedded subject may be only an overt NP ?

(120) We believed him to be the headmaster

(121) I assumed him to be innocent

(122) *We believed to be the headmaster

(123) *I assumed to be innocent

An obvious approach would be to say that the embedding verb selects a complementizer just like the complementizer *try* selects, except that it has accusative features rather than a [null] case feature:

(124) believe, [V, *uC*[acc]]

This complementizer will be specified for an [acc] feature, leading to the obligatoriness of an overt embedded subject. The derivation for an ECM clause will then be exactly the same as for a clause headed by *for*, except the complementizer will be null:

(125) $[_{CP}$ for[acc] [DP[acc] T[~~*u*D~~]... *v*[acc] V DP]]
 0 him to see her

This analysis makes these cases completely parallel to the *for* cases, except that the complementiser is null. Examples where we have a null PRO are ruled out because the null complementizer selected by *believe* would value any pronominal as [acc], and not [null].

However, as our theory stands, we simply have no way of stating the lexical entry in (124). C-selectional features are simply uninterpretable categorial features with no further structure, just like all other syntactic features in a lexical entry. Syntactic features in a lexical entry may have the properties of strength and (un)interpretability, and they may have a value, but they do not, in our current theory, have any further structure. The proposed lexical entry in (124), however, assumes a complex feature bundle is being selected, contrary to our usual assumption.

We now have to decide if we should allow such complex feature bundles. From a theoretical perspective, it would seem that this would run counter to our drive for theoretical simplicity. It also turns out that there is an alternative analysis which does not require us to change our theory.

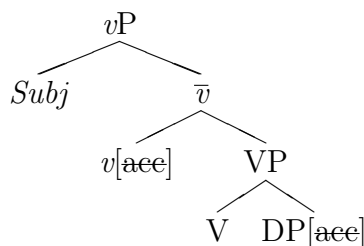
This alternative analysis is as follows: the case feature of the embedded subject is checked not by a null complementizer which transmits its feature to T, but rather by the usual case properties of the embedding verbal complex. Notice that ECM verbs may usually take a DP object:

(126) Anson believed the report.

(127) Ron expected the sack.

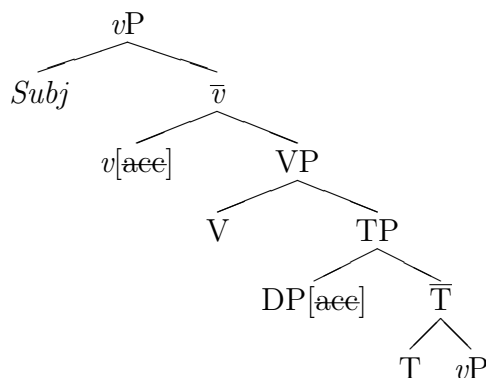
The approach we developed to the case properties of objects in general was that their accusative case was checked by an [acc] feature on little *v*.

(128)



As an alternative to the idea that ECM verbs involve a null C, we could adopt the position that they instead select a non-finite TP. The subject of this TP enters into a case checking relationship with the little *v* above *believe*, and checks [acc] in that way. This would give us a structure very similar to our normal structure for objects:

(129)



This analysis immediately explains why ECM verbs do not allow PRO:

(130) *Anson believed PRO to be happy.

Since it is little *v* that values the case feature of the lower subject, that lower subject can never be PRO. This is because little *v* bears [acc] rather than [null].

A second argument for adopting the TP analysis for ECM constructions is that control clauses and *for*-clauses behave similarly with respect to pseudoclefting, and they both contrast with ECM clauses:

(131) What Medea arranged was for her children to be poisoned.

(132) What Medea attempted was to poison her children.

(133) *What Medea believed was Jason to be a murderer.

This set of data shows that control clauses pattern in the same way as *for*-clauses with respect to pseudoclefting, but that it is impossible to pseudocleft an ECM clause. If ECM clauses and control clauses are just exactly the same (that is, they both consist of a null C which checks case on PRO or a DP), then they should pattern in the same way. However, if control clauses are CPs but ECM clauses are TPs, then we can explain the pattern we see by the assumption that it is impossible to pseudocleft a TP. This assumption is backed up by the following data:

(134) What I said was [_{CP} that we would go].(135) *What I said that was [_{TP} we would go].

If we accept these arguments then we have the following distinctions:

(136)

1. Control clauses are headed by a null C which bears [null] case. The [null] feature on C values a [case] feature on a pronominal as [null], thus forcing those verbs which obligatorily select this null C to occur with PRO, rather than an overt DP.
2. ECM clauses are just TPs, selected by higher verbs. The accusative feature of the higher little *v* checks an accusative feature on a subject in the ECM clause. PRO is ruled out because any pronoun in this position would have its case feature valued as [acc] rather than as [null].
3. *for* clauses occur with overt accusative subjects because the complementizer *for* is specified with [acc]. PRO is again ruled out because any pronoun in this position would have its case feature valued as [acc] rather than as [null].

This last point might be challenged by data from non-standard dialects of English. In certain dialects of English (those spoken in Belfast, Dundee among other places), we find examples like the following:

- (137) I went to the shop *for* to get bread.
 (138) They are going home *for* to see their parents.

However, the element *for* in these dialects seems to not be exactly the same as the complementizer *for* seen in standard English. Unlike the complementizer, the dialectal *for* may appear with verbs which do not usually accept the *for* complementizer. The following sentences are well formed in Belfast English (see the references at the back of the chapter for sources):

- (139) I tried *for* to get them.

Moreover, in ECM sentences, these dialects permit *for* but *after* the embedded subject:

- (140) I wanted Jimmy *for* to come with me.

It seems, then, that the syntactic position of this element is lower down in the clause structure than the position of the standard complementizer. We might analyse it as attached to T in some fashion, and as lacking the accusative case feature associated with the standard English complementizer *for*.

Before we leave this classification, we should note a weakness in our analysis: we need to be able to say that ECM verbs select either a finite CP, or a non-finite TP. As far as s-selection is concerned, this is unproblematic, since CPs and TPs are both propositions. But if we say that ECM verbs have a c-selectional C-feature then we need to ask why they can occur with a TP, and, vice versa, if they have a *u*T feature, why they can occur with a CP. For our purposes here, we will simply say that this is what is ‘exceptional’ about these verbs: that they allow either T or C as their c-selectional feature. This stipulation, however, is really just a ‘placeholder’ for a real analysis.

Note that the crucial theoretical ideas that we used in motivating our analysis of control clauses (and ECM clauses) came from θ -theory. In all of the examples we have been looking at, a θ -role is assigned to the subject of the embedded clause, and the whole embedded clause

is assigned a θ -role by the higher predicate. The fact that control clauses are semantically dependent on their embedding clauses (inasmuch as PRO is controlled by a higher argument) arises through a separate semantic process from θ -role assignment. In the next section we turn to a new set of cases where the same θ -theoretic principles suggest a different analysis.

8.2.6 Raising

We now turn to a further type of non-finite clause. This type occurs with certain single argument verbs, notably *seem* and *appear*, which allow non-finite as well as finite clausal arguments:

- (141) It seems that Agamemnon is a maniac
- (142) It appears that Hephaestus owns a dragon
- (143) Agamemnon seems to be a maniac
- (144) Hephaestus appears to own a dragon

Examples (141) and (142) have exactly the same thematic roles assigned as (143) and (144), respectively. In both cases the matrix verb is assigning a θ -role to a proposition. In (141) and (142) the proposition assigned the θ -role appears to be a clausal argument of the matrix verb, just as we have seen in all the cases above. In (143) and (144), however, there is no obvious constituent that receives this θ -role. Agamemnon isn't engaged in some act or state of 'seeming'. Rather the sentence is just interpreted as meaning that the situation of Agamemnon being a maniac seems to be the case.

If we were to ignore this intuition, and to blindly follow the procedure we established above, we might posit a PRO in the subject positions of the non-finite clauses in (143) and (144). After all, they look rather similar on the surface to familiar examples like (145):

- (145) Agamemnon attempted to behave well.

However, this analysis is not motivated by our principles of θ -role assignment, since the argument in the specifier of each matrix clause (*Agamemnon* and *Hephaestus* in (143) and (144) respectively) would not receive a θ -role. This means that the following kinds of structure are ruled out:

- (146) *Agamemnon seems [PRO to be a maniac]
- (147) *Hephaestus appears [PRO to own a dragon]

In (146) and (147), PRO would check a null case feature in the embedded non-finite CP, and would receive the agent θ -role of the embedded predicate. Since *seem* and *appear* are single argument verbs, they only have one θ -role to assign, and they assign it to the non-finite clause. This means that *Hephaestus* and *Agamemnon* are not assigned θ -roles. Since these DPs cannot have an expletive interpretation, the structures are ill-formed.

What then is the correct analysis of these structures? Clearly something must fill the specifier of the embedded finite T, and in order to capture the generalisation that (141) means (143) and (142) means (144), that something must be dependent for its interpretation on the

matrix subject. One way of dealing with this, which we have seen already at play elsewhere in the grammar, is to adopt the idea that the subjects in the matrix clauses have got there by movement, from underlying structures where they appear in their θ -positions.

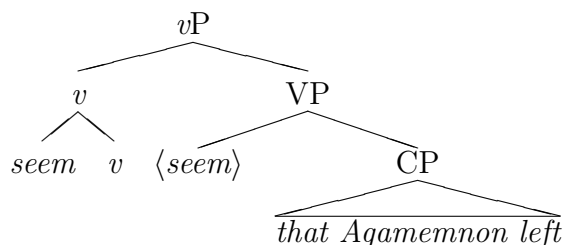
Lets see how we can make this idea more explicit.

We first start from the lexical entry for the verb *seem* (we'll concentrate on this verb, but our remarks will apply equally to *appear*, and other verbs in this class, like *turn out* and *happen*). This verb may take a single finite CP. When it does so, we have the following schematic structure:

(148) [it [seems [_{CP} that Agamemnon left]]]

Agamemnon receives its θ -role from the verb *leave* (more precisely, by being in the specifier of the little *v* associated with *leave*). It has its [nom] case checked by the finite T of the embedded clause, and T triggers movement of *Agamemnon* into its specifier to check its EPP feature. The complementizer is then Merged in, followed by the verb *seem*. This verb does not have an Agent, or any other kind of subject θ -role to assign, so we Merge an unaccusative little *v*. Recall that this was the analysis we adopted for unaccusatives in chapter 4:

(149)



We then merge T[uD, nom], as usual.

There is nothing in the structure at this point with an unchecked [nom] feature. We could raise *Agamemnon* from the embedded clause to check the EPP feature of the matrix T, but we will leave this possibility aside for the moment. We will see that such movement is ruled out by independent principles in Chapter 10:

(150) *Agamemnon seemed that left.

However, there is another possibility, which is to Merge an expletive element *it* into the specifier of *vP*, checking the [nom] feature on T with the case feature of this expletive, and raising the expletive to the specifier of TP to check the EPP feature of T and derive (149). Since expletives do not receive θ -roles, this is a legitimate operation, and results in the following sentence:

(151) It seems that Agamemnon left.

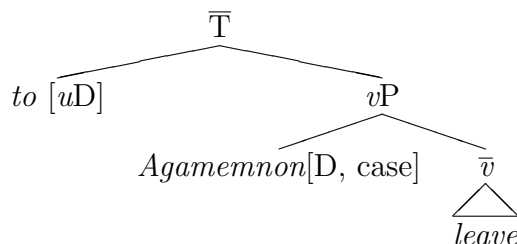
This approach derives the simple finite complement structure for *seem*. However, we are also interested in deriving the case where *Agamemnon* is the subject of the matrix clause:

(152) Agamemnon seems to have left.

We will analyse (152) by adopting the following assumption: verbs like *seem*, as well as being able to Merge with a CP, can also Merge with a TP. In a sense, then, these verbs are very much like ECM verbs in that they have the same exceptional property. Lets see, then, what happens if we choose a non-finite T in the embedded clause.

After we have built the *vP*, we have the following structure:

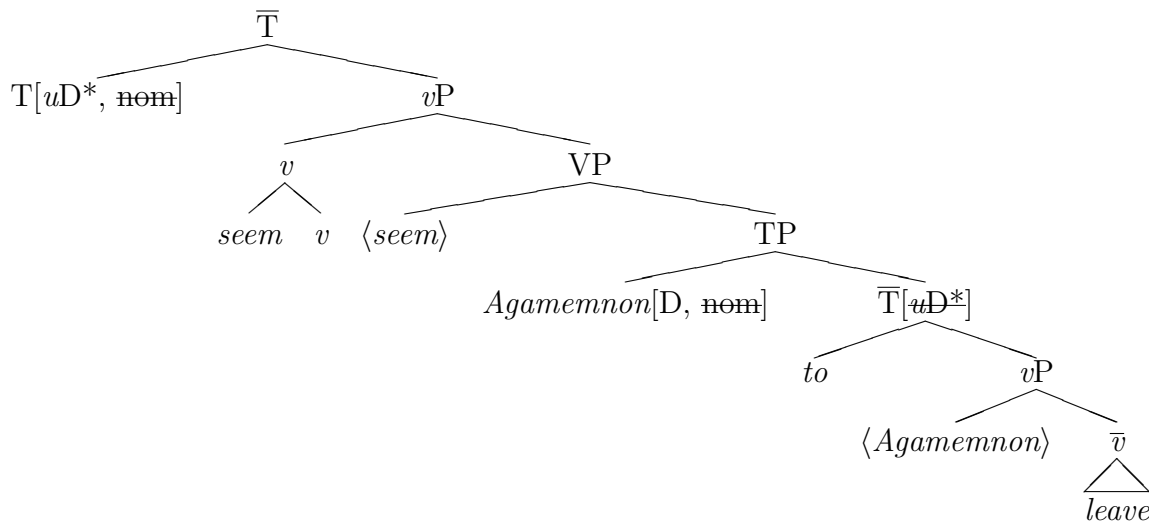
(153)



Although *to* can check its EPP feature by attracting *Agamemnon*, it cannot value *Agamemnon*'s case feature, since it lacks a case feature of its own.

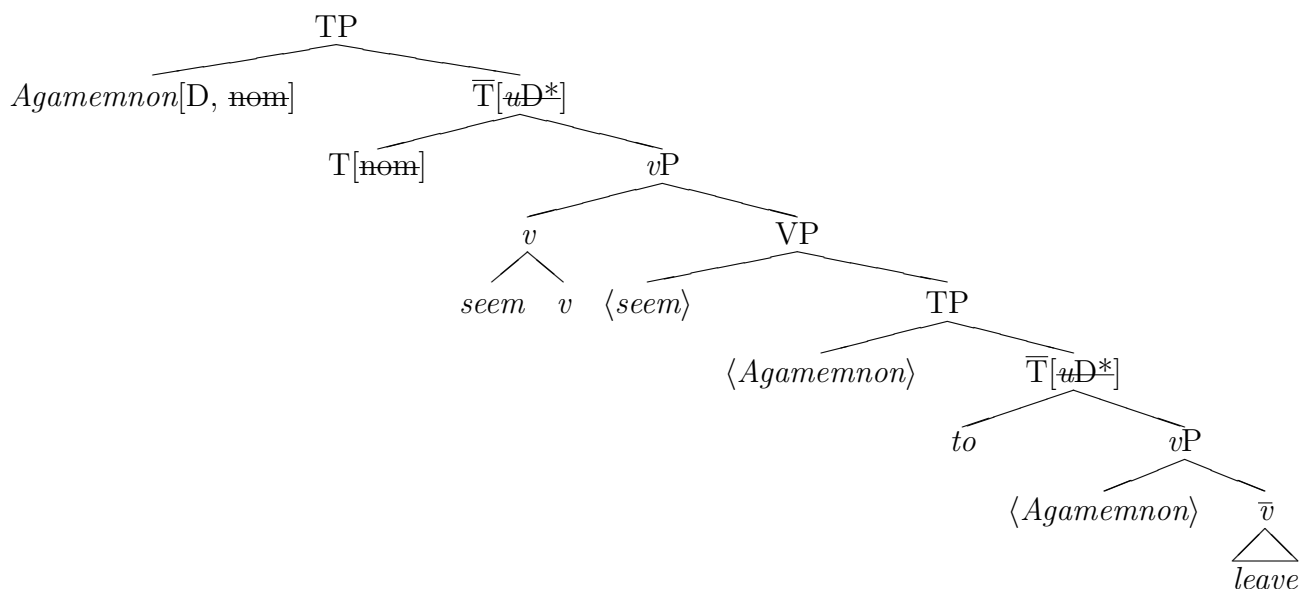
However, given that we then merge in *seem* and the higher finite T, it becomes possible to check the case feature of the matrix T with that of *Agamemnon*:

(154)



At this point in the derivation, it is also possible to move *Agamemnon* into the specifier of the higher T, since the D feature on *Agamemnon* is interpretable, and therefore has not been deleted when it was used to check the EPP feature of non-finite T. We therefore derive the following structure:

(155)



The classical name for this construction is **Subject-to-Subject Raising** (or often just **Raising**), for obvious reasons:

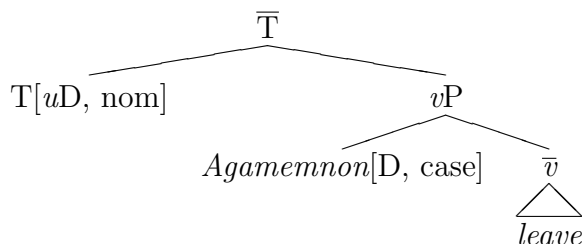
Notice that what we have here are two movements, both essentially of the same type (i.e. both driven by the EPP feature of T), both of the same syntactic formative. Firstly, the subject moves from the specifier of little *v* to the specifier of the embedded non-finite T, driven by the EPP feature of this T. Following this, the subject moves from this position to the specifier of the matrix T, again driven by the EPP property of T.

Interlude: Solving a Technical Problem

It is at this point, however, that our system of case checking faces the technical problem mentioned in Chapter 6 section 6.3.1. We have said that *seem* may select a TP, and we saw that we derived raising constructions rather neatly if that TP was non-finite. Unfortunately, if we choose a finite TP, rather than a non-finite one, then the derivation incorrectly converges, and we make the wrong prediction.

Recall that finite T bears a [nom] feature:

(156)



The [nom] feature on finite T can, under the system we have developed so far, Agree with the [~~nom~~] feature on *Agamemnon*. Remember that the non-abbreviated versions of these features are as follows:

- (157) a. T bears [*ucase:nom*]
 b. *Agamemnon* bears [~~*ucase:nom*~~]

Although the case feature of *Agamemnon* has been checked, it still Agrees with the case feature of T. Agree requires only that the features match, and we developed an analysis of subject-verb agreement which allowed a higher feature to be checked and valued by a checked lower feature. However, this approach gets us into trouble here, since we should now just be able to check the case features by matching them, and then raise *Agamemnon* satisfying EPP of the higher clause and deriving (158), with *seem* having a finite TP complement:

- (158) **Agamemnon* seems [_{TP} ⟨*Agamemnon*⟩ has left].

There are a number of solutions to this problem. A standard approach in the literature is to say that there's something special about case features, such that, when a case feature has been checked and valued, it deletes immediately. This would mean that, in our derivation above, once *Agamemnon* has checked case with the lower finite T, its case feature disappears and is no longer available to check the case feature of the higher finite T. This will mean that the derivation will crash. This approach does, however, require us to say that case features and other feature have very different syntactic behaviour, which must be stipulated somewhere.

However, another solution is available to us here, which relies on the idea we have independently made use of: that finite T bears an uninterpretable [*uclause-type:*] feature, which receives its value from C. If *seem* selects a finite TP, then the [*uclause-type:*] feature of this T must be checked and valued. However, it can't be checked by its own C, since, by hypothesis there is no such C:

- (159) seem [_{TP} ... T[*uclause-type:*] ...]

The only possibility is that it could be checked by a higher matching feature. There are two candidates for this: the higher T and the higher C. Let's see what happens when we continue the derivation:

- (160) T[*uclause-type:*] ... seem [_{TP} ... T[*uclause-type:*] ...]

At this point, the two features match, but clause type is not a privative feature: it requires a value, and no value is forthcoming. No checking is possible yet. We then continue the derivation:

- (161) C[Decl] T[~~*uclause-type: Decl*~~] ... seem [_{TP} ... T[*uclause-type:*] ...]

Now [Decl] on C has checked and valued [*uclause-type*] on the higher T. Notice by Locality of Feature Matching, it can't check the [*uclause-type:*] feature on the lower T. Also, the valued feature on the matrix T can't check the one on the lower T, since feature checking operations always have to involve the root projection of the derivation. This latter constraint follows from the Extension Condition discussed in Chapter 4, and repeated here:

- (162) **The Extension Condition:** A syntactic derivation can only be continued by applying operations to the root of the tree.

It follows that the independently needed clause-type feature, which only appears on finite T, interacts with our approach to case features to block a derivation where a subject raises from a finite TP. In fact, this approach blocks a verb like *seem* (or an ECM verb like *believe* from taking a finite TP complement. Clausal complements must either be CPs, or non-finite TPs.

Back to Raising

Raising constructions can be strung together. We find examples like the following:

(163) Hephaestus appears to have turned out to have left.

(164) Jason happens to appear to seem to be sick.

In these examples, the intermediate non-finite TPs have EPP features but no specification for case, just as in the examples we have already seen. Schematically, we have the following structure:

(165) Jason happens $\langle Jason \rangle$ to appear $\langle Jason \rangle$ to seem $\langle Jason \rangle$ to $\langle Jason \rangle$ be sick.

Notice that not all single argument verbs which take finite clausal complements can behave like raising verbs. So alongside (166), we cannot have (167):

(166) It stinks that Zeus is omnipotent.

(167) *Zeus stinks to be omnipotent.

The obvious way to deal with this is to assume that such verbs, in contrast to true raising verbs, cannot select a TP.

Before leaving raising verbs it is important to note one final structure which they participate in. As well as allowing *it*-expletives when they take a finite clausal complement, raising verbs also allow a *there*-expletive when they take a non-finite clausal complement:

(168) There seemed to be three men in the garden.

(169) There appears to be a problem with this solution.

The derivation of these sentences is fairly straightforward, although it raises an important point about the behaviour of Merge and Move. Take (168) as an example. We will assume that the verb *be* in these examples is a main verb, and takes a PP complement, and a DP subject (this is a highly controversial assumption, but since we are not focussing on the behaviour of *be* here, it will serve us just now).

(170) [[_{DP} three men] [be [_{PP} in the garden]]]

We then merge an unaccusative little *v* which attracts *be*. Then we Merge in *to*:

(171) to be[[_{DP} three men] [[_{PP} in the garden]]]

At this point, *to* has an EPP feature to be satisfied. We can satisfy this by merging in the expletive *there*. On the assumption that the *there* expletive is distinguished from the *it*

expletive by the fact that it does not have a [case] feature, it can Merge directly with the projection of non-finite (caseless) T:

(172) there to be[[_{DP} three men] [[_{PP} in the garden]]]

We then continue the derivation as above. We Merge in *seem* and its little *v* and finite T.

(173) T seem there to be[[_{DP} three men] [[_{PP} in the garden]]]

The EPP feature of finite T attracts the expletive *there*, while its case feature values the case feature on *three men*. This derives the correct result:

(174) There T seem ⟨*there*⟩ to be[[_{DP[nom]} three men] [[_{PP} in the garden]]]

Notice that the checking relation that holds between T and the embedded subject also allows us to capture the fact that the finite verb agrees with the embedded subject (that is, in this sentence we have *seem* rather than *seems*, the verb showing agreement with the plural *three men*). The agreement relation is just checking of ϕ -features between finite T and the embedded subject; the relation is structurally just the same as the case checking relation.

Although this derivation gives us the right result, it raises a new problem. At the point in the derivation where we are satisfying the EPP feature of the lower (non-finite) T, why do we not raise the subject, giving (175)?

(175) [_{DP} three men] to be[*e* [[_{PP} in the garden]]]

This is apparently a perfectly legitimate step; nothing we have said so far rules it out. Continuing this derivation gives us then the following result:

(176) *There seem three men to be in the garden.

Here we have Merged *there* with the matrix T, satisfying its EPP feature. The [nom] feature of matrix T checks the [nom] feature of the embedded subject *three men*. The EPP feature of the lower T has been satisfied by raising of *three men*.

We need some way to rule out this derivation. One solution that has been suggested is that the operation Merge is preferred over the operation Move. If we can apply either, then we take the option of Merge.

How does this help us? At the crucial step of the derivation we have the choice of Merging an expletive, or of Moving the embedded subject. Give the Merge over Move preference, only the former is possible, and so the derivation which leads to the ungrammatical outcome in (176) does not occur.

This general economy condition (if there's a choice, take Merge in preference to Move) has a range of subtle effects throughout the grammar which we will not consider here. See the reference section for further reading.

8.2.7 Object Control

Turning now to 3-place predicates with clausal arguments, we find examples like the following:

(177) Jason persuaded Medea to desert her family.

(178) Agamemnon forced Zeus to leave the school.

Looking at the θ structure of these examples, it is fairly clear that *persuade* and *force* take three rather than two arguments. This means that the structure of (177) must be (179) rather than (180):

(179) Jason persuaded [Medea] [to desert her family]

(180) Jason persuaded [Medea to desert her family]

In (179) *persuade* has the three arguments *Jason*, *Medea* and *to desert her family*, while in (180) there are only two arguments, *Jason* and *Medea to desert her family*. The structure in (180) is just the same structure that we gave to ECM verbs.

As well as this intuitive idea about θ -role assignment, we can make fairly solid syntactic arguments which show that verbs like *persuade* have a different structure from ECM verbs. The first comes from the behaviour of expletives.

Recall that expletives do not receive a θ -role. This means that they are Merged in a non- θ position, and this Merge is triggered by an EPP feature. The EPP feature appears only on T in English, which means that expletives are restricted to the specifier of TP.

This appears to be the right result. We cannot Merge a *there* expletive as an object:

(181) *I found there.

(182) *I destroyed there.

These sentences are ungrammatical because they require an object. The expletive *there* cannot behave like an object. Contrast this with the following:

(183) There seems to be a problem.

As we saw in the previous section, we can Merge an expletive in a subject position (or Move it to such a position).

With this in mind, consider now the following difference between ECM verbs and verbs in the *persuade* class:

(184) I expected there to be a problem.

(185) *I persuaded there to be a problem.

(186) I believed there to be three books on the subject.

(187) *I ordered there to be three books on the subject.

If the structures for ECM verbs and for *persuade*-class verbs were exactly the same, we would expect there to be no contrast, contrary to fact. However, if the postverbal DP in *persuade* constructions is an object, rather than an embedded subject, the contrasts we just noticed are predicted by the restrictions on the distribution of expletives.

A second argument comes from the differential behaviour of these classes of verbs when the complement is passivized. If we take an ECM verb, like (188), and then we passivize its complement, giving (189), the resulting interpretation is roughly the same:

(188) Jason expected the doctor to treat Medea

(189) Jason expected Medea to be treated by the doctor

In both cases the sentences roughly paraphrase each other: Jason has an expectation, and that expectation is that the doctor will treat Medea. Contrast this to what happens with *persuade*-class verbs:

(190) Jason persuaded the doctor to treat Medea

(191) Jason persuaded Medea to be treated by the doctor

In this case the paraphrase relation breaks down spectacularly: in (190) the doctor is being persuaded, while in (191) it is Medea that is being persuaded. This difference is again explained if the postverbal DP in an ECM construction is a subject, whereas it is an object in *persuade* constructions. This structural difference means that in (191), *Medea* is not really a passivized subject, it is just a normal object. In (190), it is *the doctor* that is the object. Since the two sentences have different object, we clearly do not expect them to be paraphrases of each other.

Contrast this with the ECM cases. On our assumption that the complement of *expect* is a full clause, passive takes place internal to this clause. This means that *expect* has the same complement in both cases, it's just that one of these is an active sentence, and the other is passive. Since actives and passives are rough paraphrases of each other, we predict the observed paraphrase relations.

These kinds of syntactic arguments back up the idea that the structure of *persuade*-class verbs is different from ECM verbs, and that the DP immediately after the verb is like a true object in the former case, while it is like a subject in the latter case. This pushes us into adopting the following ditransitive analysis for these verbs:

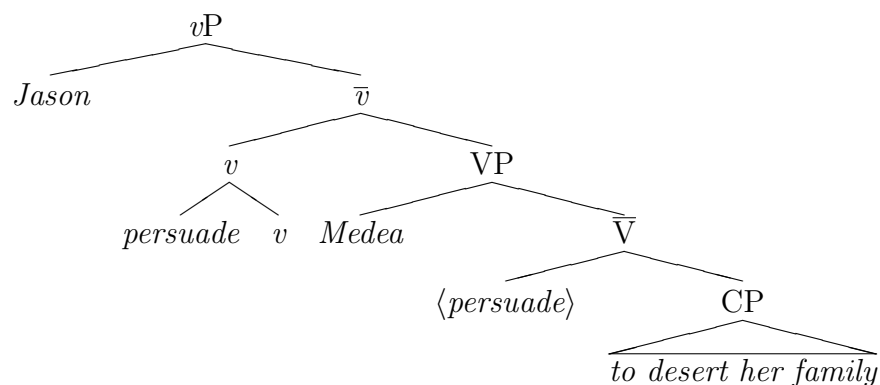
(192) Jason persuaded [Medea] [to desert her family]

This analysis is further backed up for those verbs in this class that allow a finite CP to appear in place of the non-finite one we have been considering:

(193) Jason persuaded [Medea] [that she should desert her family]

Given that these verbs are essentially ditransitive, we assume the following kind of structure:

(194)



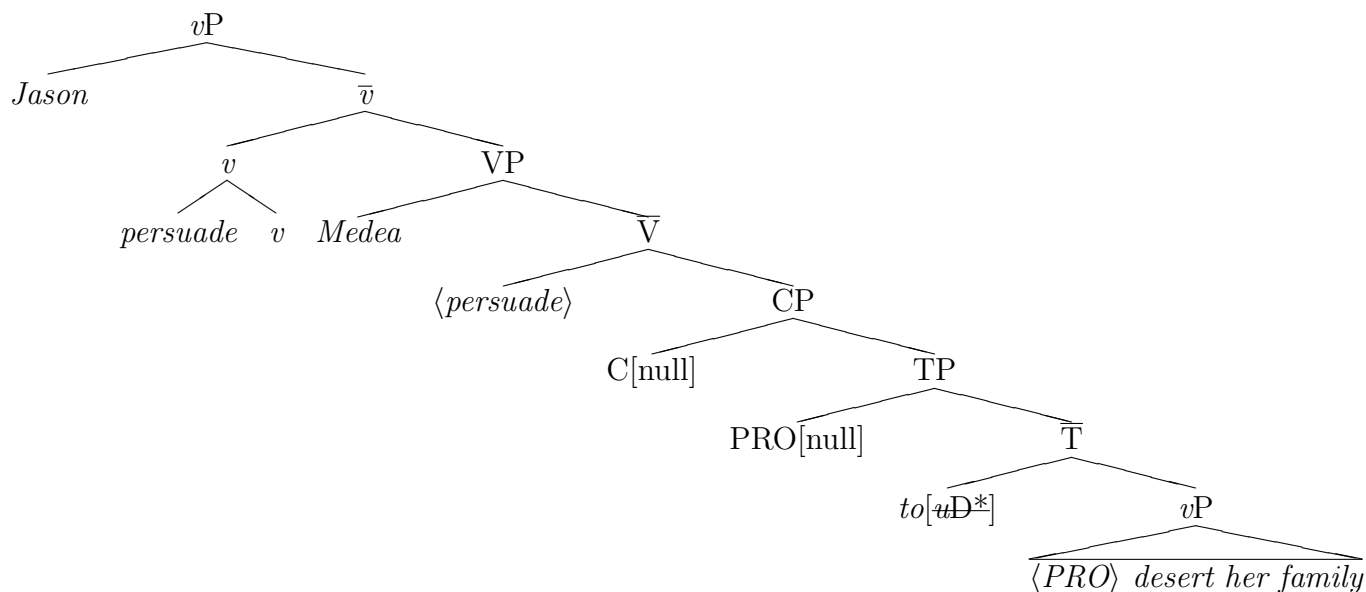
vP here will be the complement of T, and Jason will raise to the specifier of TP to check EPP and case features.

The next question is what the analysis of the embedded CP is, and more particularly, what the status of the subject of this clause is. This question can be answered immediately by reference to the θ -criterion. We know that *cut* is a two place predicate, so a θ -role has to be assigned to the subject of the embedded clause. The interpretation of this embedded subject is controlled by the matrix object (i.e. *Medea*), since it is Medea that is being interpreted as at least intending to desert her family. Since the DP *Medea* receives a θ -role from *persuade*, and since the θ -criterion prohibits a single DP from receiving more than one θ -role, it follows that *Medea* cannot have moved to its surface position from the embedded subject position. If it had, then it would be receiving θ -roles from the verb in the embedded clause as well as the verb in the matrix clause. It therefore follows that the subject of the embedded clause is PRO, rather than the trace left by raising:

(195) Jason persuaded Medea [PRO to desert her family].

Given that we have a control clause here, we extend out earlier analysis of control clauses to this case:

(196)



Since it is the object, rather than the subject, that controls the interpretation of PRO, verbs like *persuade* and *force* are said to be **object control** predicates.

8.2.8 Summary and further implications

This brings to a close our discussion of non-finite CP complementation. We have adopted the basic theoretical position that non-finite T is specified with an EPP feature but not with a case feature. The case of the subject of a non-finite clause is then determined by the

complementizer that introduces that clause (a null C[null] or *for*[acc]) or by some element outside the clause (*v*[acc] for ECM clauses and T[nom] for raising clauses).

Notice that we have now developed the theoretical technology to answer the question that we were left with at the end of section 8.1.3. We had proposed that initial CPs were in the specifier of a C head, and that there was a null pronominal in the specifier of TP, as follows:

(197) [That the world is round] *e* is obvious.

This leads to the question of why we can't simply have the following:

(198) *e* is obvious.

The answer to this is now clear: the C head that selects for an initial CP bears a [null] case feature.

8.3 Adjunct Clauses

We close this chapter with a brief look at adjunct clauses. We have already seen examples of PP and AP adjuncts, which serve to semantically modify an event or entity. We have assumed that, syntactically, these constituents are integrated into the tree via the operation Adjoin, rather than by Merge. Adjoin is not triggered by the need to check features, and it is therefore optional. CPs also may be adjuncts, which serve to semantically modify the proposition expressed by the main clause.

Adjunct CPs fall into two main classes: those that relate the propositions expressed by the two clauses in terms of time, and those that link the clauses via a relation of cause or reason.

Examples of the first class are:

(199) Before the executioner arrived, Hephaestus had escaped

(200) When the executioner arrived, Hephaestus was asleep

(201) After the executioner left, Hephaestus wept.

The adjunct clauses here are the ones introduced by the temporal complementizers *before*, *when* and *after*. Note that they can be adjoined on either side of the main clause:

(202) Hephaestus had escaped, before the executioner arrived.

(203) Hephaestus was asleep, when the executioner arrived.

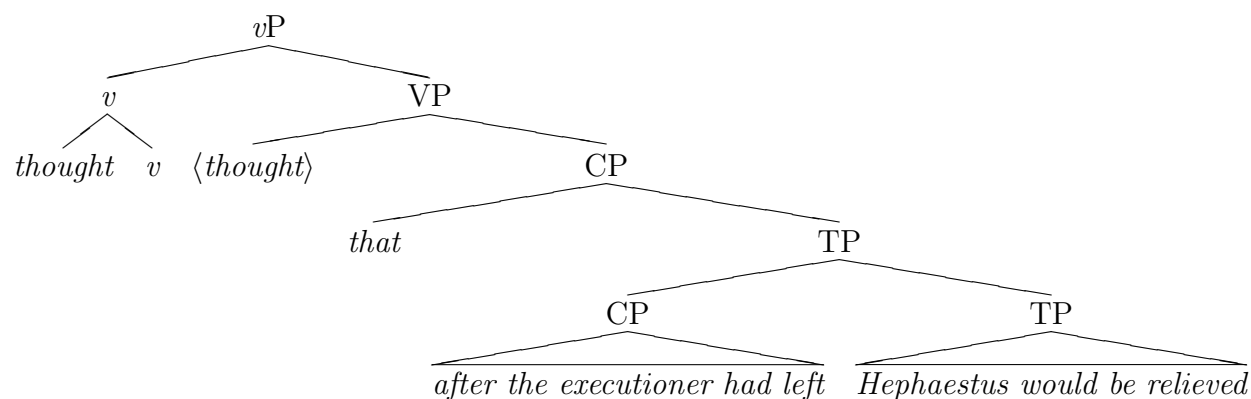
(204) Hephaestus wept, after the executioner left.

and that they appear in between embedding complementisers and the subject:

(205) Medea thought that, after the executioner had left, Hephaestus would be relieved.

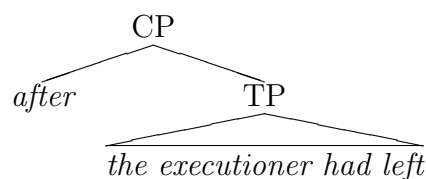
These properties suggest that temporal adjunct clauses adjoin to TP:

(206)



The internal structure of these clauses is fairly straightforward. They are headed by the temporal complementiser, which takes a *TP* complement:

(207)



The syntax of cause/reason clausal adjuncts differs little from this. These adjuncts are exemplified by clauses headed by *because*, *since* and *so that*:

(208) Medea was happy, because she had got the highest marks

(209) Because she had got the highest marks, Medea was happy

(210) Jason became invisible, so that he could escape

(211) So that he could escape, Jason became invisible

Once again, these adjuncts attach to *TP*, on either side. One problem we will not address here is the position of *so* in *so that*. The natural assumption is that it is adjoined to the complementizer *that*, but we will just assume that this apparently two word phrase is a single syntactic formative.

8.4 Cross-Linguistic Implications

In this chapter, so far, we have focussed on how the C-domain of the clause is implicated in constructing questions (via movement of *T* to *C*) and in embedded clause structure. In this section, we will look at a case where the C-domain of the clause is used in Declarative matrix constructions.

The core phenomenon we will focus on is known as **Verb-Second** (usually abbreviated as **V2**), and is found in most of the Germanic languages. The phenomenon is known as V2,

since the finite verb ends up in the second position in the clause, immediately after the first constituent. Here are some examples from German. I have italicized the first constituent, and put the finite verb in bold face:

- (212) *Ich* **las** schon letztes Jahr diesen Roman
 I read already last year this book
 ‘I read this book already last year.’
- (213) *diesen Roman* **las** Ich schon letztes Jahr
 this book read I already last year
 ‘I read this book already last year.’
- (214) *schon letztes Jahr* **las** Ich diesen Roman
 this book read I already last year
 ‘I read this book already last year.’

As you can see, the basic (θ -related) meaning of the sentence remains the same; there is a difference in meaning between these clauses, but it is related to the way that the information is presented, rather than to differences in argument structure. Roughly, we can say that the first constituent in each of these cases counts as the **topic** of the clause.

More importantly, for our purposes here, we see that some constituent appears in the first position in the sentence, and the verb appears in the second position.

We can adopt the techniques we have already devised to provide a rather elegant analysis of this phenomenon: whereas in English we proposed that when the features of C valued the [*u*clause-type] feature on T, the Q value was strong but the Decl value was weak, we can say that in German, matrix C, when it values [*u*clause-type] on T, always provides a strong feature.

Schematically, we have the following, at the point in the derivation where we Merge in a declarative C. It values the [*u*clause-type] feature as strong Decl (we adopt the usual abbreviations here, so [Decl*] stands for [*u*clause-type:Decl*]):

- (215) C[Decl] Subj T[Decl*] Adv Obj

Because of this, T (and whatever is in T) raises to C in matrix declarative clauses:

- (216) T[Decl*]+C[Decl] Subj <T> Adv Obj

However, this is only half the story. We still need to ensure that a constituent appears in the first position. Since this constituent is a topic, we will propose that any constituent can bear a [top] feature, which is eventually interpreted as signalling that that constituent is to be interpreted as a topic. In order to ensure that the topic appears locally to C, we will assume that matrix C in German bears a strong topic feature which needs to be checked: [*u*top*]

If we take our schematic structure in (216), we have the following possibilities at least, depending on whether the subject, object or adjunct is the topic:

- (217) Subj[top] T[Decl*]+C[Decl, ~~*u*top*~~] <Subj> <T[Decl*]> Adv Obj
- (218) Obj[top] T[Decl*]+C[Decl, ~~*u*top*~~] Subj <T[Decl*]> Adv <Obj>

- (219) Adv[top] T[Decl*]+C[Decl, ~~u~~top*] Subj ⟨T[Decl*]⟩ ⟨Adv⟩ Obj

Of course these are exactly the possibilities that we find in the German examples from above.

This approach is backed up by what happens in subordinate clauses in German. These clauses are, for most verbs, headed by a normal embedding C, just like English *that*. This embedding C, unlike matrix C, does not trigger V2:

- (220) Er sagte dass ich schon letztes Jahr diesen Roman **las**
 He said that I already last year this book read
 ‘He said that I read this book already last year.’

What we see here is that the verb appears to the right of the object in the embedded clause, suggesting that German is really an OV language underlyingly. In fact, if we put in an auxiliary in this sentence, we can see that the VP occurs before T as well:

- (221) Er sagte dass ich schon letztes Jahr [_{VP} diesen Roman **gelesen**] habe
 He said that I already last year this book read[past participle] have
 ‘He said that I have read this book already last year.’

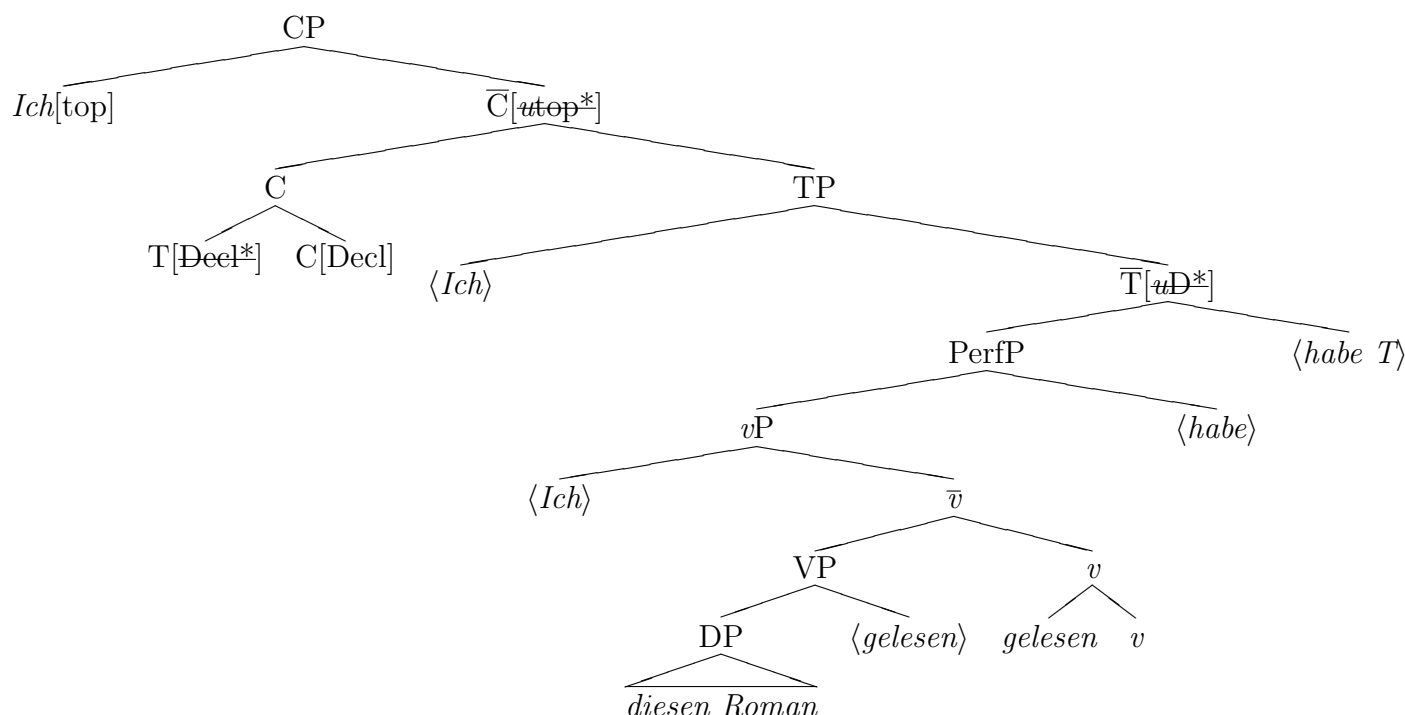
On the assumption that V raises to T in German just like in French, we can see that if T raises to C in a matrix clause, the finite verb will be in C. Since matrix C bears a Generalized EPP feature, this gives rise to V2.

This approach makes exactly the right predictions if we look at a matrix clause with an auxiliary verb. Since the auxiliary verb will be in T, and the main verb will be lower in the structure, only the auxiliary should participate in the V2 phenomenon; the main verb should stay in situ, which predicts that it will be in its base position, to the right of the object. This is exactly the right prediction:

- (222) *Ich* **habe** schon letztes Jahr diesen Roman gelesen
 I have already last year this book read[past-participle]
 ‘I have read this book already last year.’

Let’s see how this will look as a full tree structure. German is an OV language, where objects usually precede verbs, and T precedes its complement. Only C takes a complement to its right:

- (223)



Here the VP, v P and all the TP-related projections (in this case just PerfP and TP) are taken to be right-headed, with Perf raising to T, which then raises to the left-headed C. The subject *Ich* raises from the specifier of v P to the specifier of TP, to satisfy EPP, and then to the specifier of CP to satisfy the $[u_{top}^*]$ feature on C.

We have now seen a further parametric pattern: whether the $[u_{clause-type}]$ feature on T is valued as strong or weak. If it is valued as strong, then T to C raising will take place, as in English SAI constructions or German V2. Otherwise, T remains in situ. We have also proposed that German triggers movement of topics to the specifier of CP. It is not clear whether English also has such an operation. Clearly topics are able to move to an initial position in English, but, unlike in German, it does not have to be the case that something in the sentence must be such a syntactic topic:

(224) His analysis, I never liked.

(225) I never liked his analysis.

(224) shows that topics in English may appear on the left periphery of the clause, and perhaps in the specifier of CP, but (225) shows that, unlike German, this is not obligatory.

We can capture all these differences as follows:

(226)

	Infl on Aux	Infl on v	EPP on T	Decl on T	$[top]$ on C
English	strong	weak	strong	weak	optional
French	strong	strong	strong	weak	?
Swedish	weak	weak	strong	strong	strong
Scottish Gaelic	strong	strong	weak	weak	weak
German	strong	strong	strong	strong	strong

I have marked Swedish as having strong values for the Decl on T and the [top] on C parameters, since it has the V2 property, although there are many further intricacies here. Scottish Gaelic shows no evidence of T to C movement, and no evidence for syntactic topics in the specifier of CP. French does not have the V2 effect (except, as in English, for questions), and the status of obligatory topicalisation in spoken French at least, is unclear.

As usual, the more parametric options we introduce, the more urgent becomes the question of what is the status of the missing rows in the table: do these languages exist? If not, is this an accident, or is there a principled reason? Perhaps UG restricts not just principles of syntactic combination, and the architecture of the clause, but also relations between parametric values. These are empirical questions that still await an answer.

8.5 Summary

In this chapter we have examined a second major division of the clause. The first was between the structure of the projections associated with the VP and those associated with the TP. The new division we have introduced splits off the TP and the CP

We proposed that all clauses contained Cs, which implies that the Hierarchy of Projections should be expanded as follows:

- (227) **Hierarchy of Projection:**
 Clausal: C › T › (Neg) › (Perf) › (Prog) › (Pass) › *v* › V
 Nominal: D › (Poss) › *n* › N

We proposed that C is the place where the interpretable clause type feature is found. This feature determines whether a CP is interpreted as a question or as a declarative statement. This feature is also responsible for valuing an uninterpretable clause type feature on T: when this feature is strong, as in the case of Q in English, then T to C movement takes place, so that the auxiliary inverts over the subject. We abbreviated the clause-type system as follows:

- (228) a. [clause-type:Q] (interpretable clause type feature with value Q) abbreviated as [Q]
 b. [*u*clause-type:] (unchecked, unvalued uninterpretable clause type feature) abbreviated as [*u*clause-type]
 c. [*u*clause-type:Q*] (checked uninterpretable clause type feature valued as strong Q) abbreviated as [Q*].

We then looked at cases of apparent clausal subjects, and proposed that we should analyse these as involving a null pronominal element in true subject position. The apparent clausal subject is in the specifier of a special CP.

We argued that, just like T and D, C could also be a locus of case checking, and we introduced a new case feature [null] to account for the distribution of different kinds of subjects in embedded non-finite clauses. The net effect of our approach is that in non-finite clauses the subject receives its case from C or from some higher projection, while in finite clauses it has its case checked by finite T

We briefly looked at clausal modification in CPs, and proposed that these elements were adjoined to the little *n* projection. We noted that other analysis, which extend the Hierarchy of Projection yet further, are also possible for AP and PP modifiers.

Finally, we turned to questions of parametric variation, and looked at the role of the C projection in explaining the V2 effect. We then incorporated the parameters responsible for this effect into our table of parametric options.

Once again, here are the most important phenomena we met, and the basics of their explanations:

Phenomenon	Explanation
Subject Auxiliary Inversion in yes/no questions	T to C movement driven by uQ^* as a valued of [<i>u</i> clause-type] on T
Subject Clauses	In the specifier of C[<i>u</i> C, null]
Control clauses	headed by a null C[[null]]
ECM clauses	selected by Vs that can take a TP, case on subject checked by <i>v</i> [acc]
Raising Clauses	Selected by Vs that can take a TP, movement driven by EPP of higher T
V2 Phenomena	Parametric variation in the strength properties of Decl value of [<i>u</i> clause-type] on T, and [top] on C

Exercises

Exercise 1 — Obligatory *that*

Part A

Classify the following words into two groups: those that introduce their clausal complement optionally with the complementizer *that*, and those that do so obligatorily. Make up an example for each which justifies your classification. Some of the words are ambiguous in their syntactic category, and I have indicated whether you should include both possibilities, or just the verbal case.

belief; believe; sure; important; indicate; indication; signal (V and N); shout (verb only); scribble down; claim (V and N); whisper; report (V and N); murmur (V and N); fax (V only); email (V only); certain; amazing; posit; exclaim; exclamation.

Part B

What generalisations can you draw from your classification? Think about both syntactic and semantic possibilities.

Exercise 2 — Negation in Interrogatives

Assume the analysis of Subject Auxiliary Inversion given in the chapter. Explain what problems arise for this analysis given the well-formedness of examples like (1) and (2):

- (1) Hasn't the potion worked?
- (2) Didn't the magic work?

Hypothesise a way to get round this problem, and outline an analysis. Explain what you think the strengths and weaknesses of your analysis are. Hint: concentrate on the different behaviour of the two expressions of negation, *n't* and *not*. You might find the following data helpful:

- (3) Has the potion not worked?
- (4) *Has not the potion worked

Exercise 3 — Embedded Inversion

In some dialects of English, notably many of those spoken in Ireland, we find so called **embedded inversion**. These are structures where SAI occurs in embedded clauses, selected by an interrogative predicate:

- (5) He asked did I live there.
- (6) I wondered had he left yet.

What do you think is going on here? Draw trees for these sentences under your hypothesis.

Exercise 4 — Combining verbal classes

The following sentences involve at least two degrees of embedding. Discuss the derivation of each, saying how the mechanisms for control, ECM and raising interact.

- (7) Medea tended to appear to be evil.
- (8) Jason persuaded Medea to try to run away.
- (9) Agamemnon expected Clytemnestra to seem to be happy.
- (10) Zeus wanted Hera to persuade Athena to leave.
- (11) For Zeus to appear to be happy would be impossible.
- (12) Hera tried to appear to be happy.

Exercise 5 — Trees for Prefer.

Draw trees for the following three sentences, and then discuss what these structures imply for the lexical entry for *prefer*.

- (13) Jason would prefer for Medea to have cursed Agamemnon.

- (14) Jason would prefer Medea to have cursed Agamemnon.
 (15) Jason would prefer to have cursed Agamemnon.

Exercise 6 — Other non-finite clausal complements

Consider the following sentences. Assume that T in each case contains the inflectional affix *-ing*. Try to motivate analyses which maximise the parallelism between these constructions and the non-finite constructions discussed in this chapter:

- (16) Jason refrained from casting the spell
 (17) Agamemnon stopped Jason casting the spell
 (18) Agamemnon stopped Jason from casting the spell
 (19) Jason thought of defending the dragon
 (20) Medea denied poisoning the phoenix.

Exercise 7 — Classifying verbs

Use the tests discussed in the text (expletives, passivisation etc) to determine whether the following verbs are raising, ECM, subject control or object control. We are only interested in cases where the verb is followed by an infinitive headed by *to*:

nominate; allege; command; fail; guarantee; judge; happen; convince; choose; demonstrate; pick; remain; know

Exercise 8 — V2 trees

Draw a tree for the following sentence of German, marking in all movements, and indicating the relevant checking relationships (I have altered standard German punctuation conventions here in case this leads to confusion):

- (21) Er sagte dass er gestern schon angekommen ist.
 He said that he yesterday already arrived is
 ‘He said that he had arrived already yesterday.’

Now look at the following example, where the complementizer *dass* is missing. This is a case of **embedded V2**. Provide an analysis for it and draw a tree structure embodying your analysis:

- (22) Er sagte gestern ist er schon angekommen.
 He said yesterday is he already arrived
 ‘He said that he had arrived already yesterday.’

Exercise 9 — Trees once again!

This exercise involves putting together everything you have learned in this chapter. Draw tree structures for the following sentences, and annotate each tree with a discussion of the

motivation of any phonologically null categories (PRO, trace, null complementizers, etc.):

- (23) Because they hated him, the druids forced Jason to live in a cupboard
- (24) The students decided that, after they had passed the test, they would take a trip on a magic carpet.
- (25) It appears that the magic potion has fallen into the wrong hands, since the mandrakes don't seem to be growing.
- (26) That Areadne tended to collapse when she was kissing the minotaur didn't surprise Theseus, because she had always been prone to fits of fainting.
- (27) Athena decided to persuade Agamemnon to forgive Jason, since she was sure that he was not likely to be lenient to him.
- (28) When dining with evil crocodiles, it is advisable to wear armour.

Exercise 10 — Locality in Binding

Recall that we have developed three generalisations about the distribution of reflexives, pronouns and R-expressions, which we abbreviated using the concept of Binding (see the exercises in chapter 4. Our latest versions of the Reflexive Generalisation is repeated here:

The Reflexive Generalisation: A reflexive must be bound.

Recall that for A to bind B, A must c-command B and A and B must be coreferential.

Part A

It turns out that this generalisation needs to be further restricted. The following set of data makes this point. For each example, explain how the generalisation above is satisfied, and try to characterise the nature of the locality constraint:

- (1) *Anson thought that himself was happy.
- (2) *Anson believed Jenny to have hurt himself.
- (3) *Anson left before Jenny saw himself.

Part B

Some linguists have argued that the relation between the reflexive and its antecedent is similar to the feature checking relation discussed in the text. Feature checking requires c-command, as does reflexive binding. Feature checking requires that the features on both elements involved in the operation are the same, and so does binding. Finally, Feature checking is constrained so that it only looks at the closest element with matching features. Go through part A once more, and see if you can use these properties of feature checking to explain the ungrammaticality of the examples, adopting the following working hypothesis:

Reflexive Binding: The ϕ -features of a reflexive must be checked by a higher antecedent.

One of the sentences in part A is problematic for this approach. Say which, and explain why.

Explain how the following refinement predicts the correct result:

Reflexive Binding: The ϕ -features of a reflexive must be checked by a higher antecedent within the same CP.

Part C

Go through the derivation for the following examples, and explain why they are grammatical, under this set of assumptions:

- (4) Anson believed himself to be handsome.
- (5) Anson tried to shave himself.
- (6) *Jonathan persuaded Kate to lick himself.

Further Reading

Section 8.1.1: The first exploration of the properties of complementizers is (Bresnan 1970). The idea that sentences are projections of C stems from Chomsky (1986a), and has been adopted in almost all subsequent work within the general framework we are adopting in this book. See Roberts (1996), Haegeman and Guéron (1999), Ouhalla (1998).

Section 8.1.2: Grimshaw (1979) is a detailed exploration of the different syntactic and semantic selectional properties of various kinds of verbs. See Adger and Quer (2001) for a discussion of how the syntax and semantics of verbs which select [Q] CPs works, and Nakajima (1996) for further detail.

Section 8.1.3: The two analyses we compared in the text are updatings of Ross (1967) and Koster (1978).

Section 8.2: Initial work on the distribution (and interpretation) of PRO stems from Chomsky (1981). For the [null] case approach, see Chomsky and Lasnik (1993) (published also as Chapter 1 of Chomsky (1995b)). See also Koster (1984), the papers in Richard Larson and Higginbotham (1992), and for more recent approaches rejecting the θ -theoretic assumptions in the text Hornstein (1999) and Manzini and Roussou (2000). The syntax of ECM constructions was extremely important in the development of Generative Grammar; see (Postal 1974) for a book length discussion of different non-finite complementation types. The current approach is a development of Chomsky (1981), which in turn develops ideas from Rouveret and Vergnaud (1980). More recently, various authors have argued that objects in English, just like subjects, raise to the specifier of a functional projection (see, especially, Johnson 1991). For arguments based on ECM constructions see Lasnik and Saito (1991), and, more recently, Lasnik (2001). Raising constructions also received a lot of attention in early Generative Grammar (again, see Postal 1974). The interaction of raising with expletives is considered in detail in Chomsky (1995b), and Chomsky (2000).

Section 8.4: The V2 phenomenon is a traditional concern of descriptive Germanic grammarians. The approach outlined in the text stems from Besten (1983), but see Travis (1984) and Zwart (1993) for an alternative. The examples in the text are adapted from Roberts (2001) who situates V2 in a general theory of head movement. Roberts and Roussou (to appear) is a theoretical account which is close in spirit to what we discussed in the text,

although they propose that the requirement that the specifier of CP be filled is due to an EPP feature on C, rather than a feature related to topic-hood. For the syntactic position of topics, an important reference is Rizzi (1997).